REAL-TIME PRICE MONITORING AND MARKET PREDICTION FOR AGRICULTURAL COMMODITIES IN DISTRICT 2: A MULTI-PLATFORM DATA ANALYTICS APPLICATION

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In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology
Specialized in Web and Mobile Application Development

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APPROVAL SHEET

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THE RESEARCHERS

DEDICATION

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J.V.S

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ABSTRACT

The study titled “Real-Time Price Monitoring and Market Prediction for Agricultural Commodities in District 2: A Multi-Platform Data Analytics Application” was developed to improve market price monitoring in Laguna’s Second District, covering Cabuyao City, Los Baños, and Bay. The project addressed challenges caused by inflation and the lack of real-time price information, which often result in price volatility and unfair trade practices in public markets.The system provides a centralized, multi-platform solution that enables consumers, farmers, and public market stall owners to access and share up-to-date agricultural and poultry product prices. It consists of a mobile application for general users and a web-based administrative platform for data management. Prices are confidentially submitted by sellers, validated by administrators, and aggregated to compute average market prices, ensuring transparency and reducing biased or misleading information.The study followed the Agile-Scrum development methodology and integrated data analytics features for price visualization and market trend analysis. A machine learning–based prediction module was implemented to analyze historical price data and generate predictive insights on future price fluctuations, supporting informed decision-making.System performance was evaluated using an ISO-based quality assessment covering nine characteristics, including functional suitability, efficiency, usability, and security. Results indicated overall ratings of “High” to “Very High,” demonstrating the system’s effectiveness, reliability, and accuracy in price prediction.In conclusion, Price Pulse proved to be an innovative and user-friendly platform that supports fair pricing practices and strengthens the local market economy in District 2. Future enhancements are recommended through the integration of more advanced predictive and analytical models to further improve accuracy and system performance.

Keywords: *Real-time Price Monitoring, Market Prediction, Agricultural Commodities, Multi-platform Application, Machine Learning Algorithms*

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DEFINITION OF TERMS

ARCH/GARCH Models	Statistical models used to measure and predict time-varying volatility in commodity prices based on past fluctuations.
Backend System	The server-side part of an application is responsible for processing data, managing storage, and running system logic.
Commodity	A basic agricultural or poultry product that can be bought, sold, or traded in the market, such as rice, onions, eggs, and vegetables.
Data Analytics	The process of examining raw data using statistical tools and algorithms to uncover patterns, insights, and trends.
Data Validation	The procedure of verifying the accuracy, completeness, and consistency of data before it is stored or displayed in the system.
Forecast Refit	A model-updating process where new daily data is used to retrain or fine-tune a forecasting model for improved accuracy.
Localized Market Data	Market information gathered from a specific geographic region to reflect real-time, local economic conditions.
Machine Learning	A field of artificial intelligence that enables a system to learn from historical data and improve its predictions over time.
Mean Absolute Percentage Error (MAPE)	A measure of prediction accuracy that expresses forecasting errors as percentages
Multi-Platform System	Software designed to operate across various devices, such as mobile phones and desktop browsers without losing functionality.
Predictive Analytics	The use of statistical models and algorithms to forecast future events, such as commodity price changes.

Price Manipulation	An unethical act of artificially altering market prices to mislead buyers or gain unfair advantage.
Real-Time Data Synchronization	The automatic and instantaneous updating of information across all devices connected to the system.
Time Series Data	A collection of data points recorded at consistent time intervals, often used for trend analysis and forecasting.
Web-Based System	A system or application that runs on a web browser and can be accessed through the internet without installation.

CHAPTER 1

INTRODUCTION

Project Context

The Philippines experienced inflation, soaring prices of almost all essential goods, especially food, and fuel, which have been aggravating factors for national inflation persistently in recent years. These price increases affected fundamental economic households, which are either really significant or are part of the most important items, whose prices have been increasing the most-namely, key agricultural and poultry commodities. Inflation is higher than wage increases, which leads Filipino families to cut spending, with spillover effects on local markets and supply chains.

Inflation tends to breathe down the neck of local markets in Laguna, a province characterized by urbanization mixed with agriculture. Prices for important things like vegetables and poultry as well as rice tend to rise and fall such that it affects consumers and market farmers alike. Further, this scenario has added more economic pressure in areas where agriculture is still a major means of living. More importantly, it signifies the urgent need for localized data-a vital but neglected management tool for local food cost control.

In the case of Laguna's Second District, which consists of the City of Cabuyao, and the municipalities of Bay and Los Baños, residents face the direct implications of inflation, especially in commodities of basic need. Many of these important staple goods have witnessed drastic increases in local market prices, such as rice, vegetables, and poultry. This condition affects both consumers

and small farmers who all lack real-time data access that can guide them for their buying and selling decisions. Lack of real-time pricing further serves to aggravate the problems confronting these communities, thus calling for more visibility and access to market conditions.

Onions, tomatoes, rice, and poultry considering these all-too-familiar products are staples in a Filipino household. Their growth is very vulnerable to environmental calamities such as typhoons and droughts, making production fail in varied seasons in the Philippines. Poor storage and distribution systems in the country compound problems, causing seasonal spikes in prices and unstable markets.

Instability in price signals is a challenge for traders, farmers, and consumers. There is a possibility that a crop sold may be a loss to a farmer if sold in the wrong time, while high price values may have to be incurred by consumers from distant areas. Too much informality in the flow of goods and erroneous information has contributed to price manipulation and unfair trade practices, eventually deepening the inequality in economically depressed agricultural communities.

The country has been relying on the traditional methodology of price monitoring through executing regular surveys by government agencies like the Philippine Statistics Authority (PSA) and the Department of Agriculture (DA). However, reports usually come late and often turn out to be inaccurate in real-time. Despite being the best forecasting tool, research has shown that ARIMA and simple time series analysis have been used in various academic studies.

These techniques have yet to gain popularity among local government units or farming practices.

The most significant challenges facing price monitoring today are delays in reporting, errors from manual tracking, and the lack of integration. Many stakeholders still depend on word-of-mouth or outdated bulletin boards for pricing. Yet, while this has been the case, some machine learning models are promising.

This created a requirement for a price monitoring system that can be timely, accurate, and accessible to provide both consumers and producers information electronically in real time. The conditions existing at present involve late, manually operated, and fragmented data reporting systems, which restrain effective decision making at the local level-distribution areas such as those of Laguna's Second District. The researchers proposed the development of a machine learning-powered localized price monitoring platform that integrates real-time market data from key sources and provides predictive analytics for essential commodities such as rice, vegetables, and poultry. It aimed to bridge the information gap, smoothen price fluctuations, and enhance evidence-based planning by farmers, traders, and local government units. This great technological advancement promotes fair trade practices; improve food security; and realize a more resilient local economy.

Project Objectives

The main objective of this research is to develop a mobile application for real-time price monitoring and market price prediction of a supply of

agricultural and poultry products using data analytics in the Los Baños public market.

1. Develop a multi-platform application using web based and mobile based application that provides the real-time price of agricultural and poultry products.
2. Integrate data analytics features that visualize price trends, predict market fluctuations, and offer insights for better decision-making in agricultural trade.
3. Evaluate the system's performance in terms of accuracy, usability, and reliability through user testing, feedback collection, and comparative analysis with existing price monitoring methods using ISO testing.

Project Purpose

This project developed a multi-platform web and mobile application that helps monitor real-time prices and provide market predictions for agricultural products in District 2, Laguna. The system is mainly built to help small farmers, local vendors, buyers, and even LGUs by giving them easy access to updated and accurate price data. By showing the average market prices collected from different sellers, the platform hopes to improve transparency and help people make better decisions when it comes to selling or buying agricultural goods.

This project supports some of the United Nations' Sustainable Development Goals, especially SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production). For SDG 1 (No Poverty), the system helps farmers and vendors avoid being underpaid or overpricing their products

because they now have access to fair market prices. This helps improve income and avoid losses. For SDG 2 (Zero Hunger), the system supports better food distribution by helping prevent overproduction or shortages through real-time market data. SDG 9 (Industry, Innovation and Infrastructure) is supported by introducing a digital solution to a manual and outdated process, which makes the agricultural market more modern and efficient. And for SDG 12 (Responsible Consumption and Production), the project encourages smarter buying and selling by showing validated, up-to-date data that can guide both sellers and consumers in making the right choices.

The system collects prices submitted by sellers. These prices are not shown individually to the public. Instead, the admin reviews the submission, validates the data, and compute the average price of each product. This way, users only see fair, balanced prices without showing which shop submitted what.

The platform is available on both mobile and desktop, so users can check the data anytime. It also has features like price trend graphs and simple predictions to help people spot patterns and plan ahead. The project mainly focuses on areas within District 2 of Laguna - Cabuyao City, Bay, and Los Baños.

This project hopes to support local farmers and sellers, reduce unfair pricing, promote smarter trade, and help the agricultural economy in District 2 by using technology that’s simple, reliable, and useful for everyone.

Sellers. can submit the current prices of their products, contributing to a shared price pool. While individual prices remain confidential, their data helps

build a transparent and inclusive market system that reflects fair and accurate trends.

Admin. Administrators are responsible for validating all submitted data and calculating the average prices. Their role ensures that only trustworthy and accurate information is shown to users, enhancing the credibility of the platform.

Public Users (Farmers, Consumers, Traders). Users gain access to a simplified and user-friendly dashboard that shows averaged commodity prices and market trends. This allows them to make better financial and purchasing decisions without being misled by outliers or biased data.

Scope and Limitation of the Study

The project established real-time price-monitored and market-insight such as the developments of agricultural and poultry commodities in District 2 of Laguna particularly public markets in the City of Cabuyao, Municipality of Bay, and Los Baños. It shall collect, process, and present updated prices and price trends on 60 of the basic and agricultural commodities, including rice, poultry, vegetables, onions, and tomatoes-such as markets in the historical data from 2020 to 2024, and from obtaining from government agencies, from public sources and user-specified inputs.

The system develops two major modules: a mobile application targeted to end-users, such as farmers, vendors, and consumers; and a web-based administrative platform to manage, validate, and update the price data. In addition, the machine learning techniques were integrated into the use of

analyzing historical data to achieve predictive insights on market price fluctuations so that good decision-making is enhanced.

Price Submission Module. This part of the system is where registered sellers can submit the current prices of their agricultural products. They are expected to update the prices regularly, especially for common and high-demand items. These price entries become the main data used by the system for analysis and public display. However, to keep things fair and protect sellers' privacy, the individual prices they submit won't be shown to users. Instead, all submitted prices go through the admin, who checks and validates them before the system calculates the average price. This helps avoid biased or misleading information and keeps the data neutral for everyone.

Admin Validation and Data Management. The admin has an important role in making sure the submitted prices are accurate and reliable. Every entry is reviewed before it's included in the system. Once all prices are checked, the system computes the average price for each product, using data from multiple sellers. The admin also manages things like product categories, user accounts, and the content shown on the dashboard. This helps keep the system organized and ensures that the public sees only clean and verified data.

Analytics and Price Monitoring Interface. This module shows the average prices of products in a clear and easy-to-understand way. It's made for farmers, sellers, and regular buyers who want to keep track of market prices. The page includes real-time price displays, simple graphs showing trends, and comparisons between current and previous prices. It's designed to help users make smarter choices when buying or selling. People don't need to sign up or

log in just to view prices, so it's accessible for everyone who wants to stay updated with the market.

This study focuses on the development of a multi-platform application, consisting of both a web-based system and a mobile-based application, that provides real-time pricing information for agricultural and poultry products. The system allowed farmers, sellers, buyers, and market administrators to monitor current market prices efficiently and predict price trends. The mobile application caters primarily to everyday users (farmers, sellers, consumers), while the web-based platform allows administrators to manage product listings, verify user submissions, and oversee market data.

There are a number of limitations in this study. The enclosed geographical boundaries of this project are only in District 2 Laguna, which are not be accessible to parts of other districts and provinces. The data sources may be affected by gaps or inconsistencies, especially in which the market has little or irregular documentation, data collection practices. On the other side, the accuracy of the forecast relies heavily on the overall historical data set in terms of completeness and quality. For restrictions outside those such as weather anomalies, government policies, or world marketplace shifts, the prices might be out of bounds from system predictions. The first rollout focused only on selected public markets within the target district, with no provisions for inter-market comparisons or continental forecast features.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter presents a summary of existing research relevant to the study. It provides an overview of previous studies that have explored market prediction and price monitoring. By analyzing these works, this chapter establishes the foundation to the current study, demonstrating its significance and contribution to the field.

Analysis and Impact of Price Volatility

The volatility in the prices of agricultural produce is influenced by various factors such as seasonal changes, supply and demand, etc. According to Piotr et al., (2015), based on the observations that have been collected from the Ministry of Agriculture and Rural Development in Poland, the researcher's analysis has found that there was a big difference in price volatility for Polish commodities which was connected with the global crisis, with the rise of prices for agricultural commodity brought market problems such as increased prices for bread and food. The negative impact of price volatility not only limits consumers but also reaches the producers. It can create problems especially in planning production, in the whole, it can lead to loss of trading position as well as welfare loss of consumers (Kemény et al., 2012).

Considering that price volatility brings significant impact, the analysis of price volatility would be a significant feature for a price monitoring system. Knowing the type of model to use for a study is important as it affects the

accuracy of the analysis. Khabibah et al., (2019) conducted a study that analyses price volatility that happens at the level of producers and consumers of big cayenne in Malang Districts; to analyze the price volatility, the method they used is ARCH/GARCH models. Their study showed that the price volatility of big cayenne in Malang Regency at the producer and consumer level is low.

According to Calvo G. A., (2006), sudden stops are associated with increased volatility in relative prices. They introduced a model based on information acquisition to rationalize the increased volatility. They conducted an empirical analysis of the conditional variance of the wholesale price to consumer price ratio using ARCH techniques which confirm the relevance of sudden stops and potential balance sheet effects as key determinants of relative price volatility, where they captured balance sheet effects by the interaction of a proxy for potential changes in the real exchange rate.

Supriya & Mamilla (2022), highlighted that seasonality in commodity markets often arises from the imbalances of supply-demand, and this can complicate the price discovery process. The GARCH model was employed for seasonal changes, using this model the researchers can analyze and predict price movements better. Using this approach, it allows for a more nuanced understanding on how price volatility is affected by seasonal patterns.

Kamal and Bablu examine the transformative role of mobile applications in empowering smallholder farmers, particularly in developing countries. They highlight how these digital tools provide farmers with real-time access to critical agricultural information, including weather forecasts, crop prices, best farming practices, and pest management techniques. By facilitating direct connections

between farmers and buyers, mobile apps reduce reliance on intermediaries, lower transaction costs, and enhance price transparency. Additionally, the study discusses how mobile applications offer financial services such as mobile banking and digital payments, improving financial inclusion for farmers. These tools also aid in resource management by allowing farmers to monitor inputs like water and fertilizer usage, leading to increased efficiency and productivity. The research underscores the significant impact of mobile applications on agricultural development and rural livelihoods.

Application of ICT in Market Monitoring of Agricultural Products

Sudden changes in the price of agricultural products or price volatility are common in Agricultural commodities; to handle the sudden changes in the price of agricultural products, they integrated ICT into the process. S. A. A. Rajon et al., (2014), presented an application of ICT to disseminate market price information to the consumers. The consumer is charged through the identification of his mobile phone within a fair price margin through SMS, the seller also uses SMS to demand the price and at the same time confirmation is made by the buyer. The price margin should be defined by the government or legitimate agencies that are appointed to monitor the market. Knowing the price margin of a commodity it is unlikely for the consumer to buy their commodity for a higher price than the fair price margin.

Consumers purchasing behaviors and consumption of beef are linked to the retail price of meat. In the study of Charleboi et al., (2016) about retail price volatility and food consumption motivation stated that higher price has forced 37.9 percent of the sample to reduce or to completely stop consumption of beef

from the last 12 months. Changes in retail prices of agricultural commodities or meat can visibly affect the purchasing behavior of consumers toward the commodities.

Prices of poultry products in the market continued to rise but the demand for poultry products increased while the supply for poultry products has drop in output as the producers for poultry products stop production due to the rising prices of inputs and poultry disease, even with the high price and demand for poultry products, producers have stopped producing poultry products as it comes with a high risk because of increase in prices of inputs and lack of vaccine to support the prevention of poultry disease DA P. O. 2022.

Li et. al. (2024) explored the application of an improved Particle Swarm Optimization-Genetic Algorithm (PSO-GA) in forecasting the prices of agricultural products, specifically focusing on scallion prices in Shandong Province. The study revealed that the proposed semi-heterogeneous PSO-GA model significantly reduced the Mean Absolute Percentage Error (MAPE) and improved directional accuracy compared to traditional models, indicating its potential for practical applications in agricultural price forecasting.

Cayabyab and colleagues developed *Price Watch*, a mobile application designed to assist Filipino consumers by providing real-time price monitoring of agricultural products in public markets. The app utilizes the Exponential Smoothing algorithm to forecast commodity prices, allowing users to view historical price trends and anticipate future price movements. This tool enhance transparency in market pricing, enabling consumers to make informed purchasing decisions and better manage their household budgets. The

application was evaluated using ISO/IEC/IEEE 29119 Testing Standards and received a high acceptability rating from users, indicating its effectiveness and reliability in providing timely market information.

Fafchamps and Aker (2015) studied the impact of mobile phones on agricultural pricing transparency in West Africa. Their research found that farmers who had access to mobile-based market price updates were able to sell their goods at better rates and avoid middlemen who manipulated prices. The study emphasized the importance of user-generated price reporting, where farmers share price data through mobile applications, making the market more transparent and fairer.

The integration of artificial intelligence (AI) into livestock farming is revolutionizing the industry by enhancing efficiency and productivity. According to a market analysis by Market.us (2023), the AI in livestock farming market is projected to reach approximately USD 2.9 billion by 2033, with a compound annual growth rate (CAGR) of 23.8% from 2024 to 2033. This growth is driven by the adoption of AI technologies that enable real-time monitoring, predictive analytics, and automated decision-making processes in livestock management. These advancements facilitate improved animal health, optimized resource utilization, and enhanced production outcomes.

Bista and colleagues developed *Crop Mandi*, a mobile application aimed at forecasting demand and prices for agricultural crops. The app leverages data analytics to provide farmers with insights into market trends, enabling them to make informed decisions regarding crop production and sales. By predicting price movements, *Crop Mandi* helps farmers plan their planting and harvesting

schedules to align with market demand, thereby maximizing profitability. The application serves as a decision-support tool, reducing the risks associated with price volatility and enhancing the bargaining power of small-scale producers. This technological intervention demonstrates the potential of mobile applications to improve market efficiency and support sustainable agricultural practices.

Raw Products and Livestock Producer Price Index

According to Del Prado, Produce Price Survey (Philippine Statistics Authority, 2024), the Philippines' Producer Price Index (PPI) for agriculture has recorded a slower annual decline of 5.7 percent in the third quarter of 2024 from an annual drop of 7.7 percent in the second quarter of 2024. In the same quarter of the previous year, the annual growth rate of PPI for agriculture was registered at 8.6 percent. On a quarterly basis, the PPI for agriculture recorded a 2.2 percent increase during the third quarter of 2024 from 4.4 percent quarter-on-quarter decrease in the second quarter of 2024. At the national level, the annual growth rate of the PPI for crops has a slower annual decrease of 8.0 percent in the third quarter of 2024 from a 10.1 percent year-on-year decline recorded in the second quarter of 2024. In the third quarter of the previous year, its annual increment was recorded at 12.5 percent. The PPI for livestock and poultry recorded a slower annual increase of 7.1 percent during the third quarter of 2024 from 9.7 percent annual increment in the second quarter of 2024. In the third quarter of 2024, from 9.7 percent annual increment in the second quarter of 2024. In the third quarter of 2023, the PPI for livestock and poultry noted an annual decrement of 1.7 percent. Slower annual increments were observed in

the PPI for livestock at 6.7 percent in the third quarter of 2024 from 8.8 percent annual increment in the second quarter of 2024, and PPI for poultry with 7.9 percent year-on-year increase in the third quarter of 2024 from an 11.4 percent increase in the previous quarter. The PPI of livestock and poultry exhibited a downtrend as it posted 4.8 percent annual decline during the quarter from a 3.7 percent increase recorded in the second quarter of 2024. It is also the same on the PPI for fisheries that recorded a slower annual drop of 8.3 percent in the third quarter of 2024 from 14.7 percent annual decrease in the second quarter of 2024. In the third quarter of 2023, the PPI for fisheries registered a 0.9 percent annual decrease.

It is important to study the main factors that affect price fluctuations of a meat like pork, especially in China where it is an important livelihood indicator. In a study by Pang et al., 2023 where they analyze the main factor that affects the pork price fluctuations using a structural vector autoregressive (SVAR) model from the aspect of price, supply and demand changes, and pig epidemic shocks. It is indicated in the result that the pork price contributed the largest with 29.60% impact on the pork price 18 months later, while the supply factor that affected the pork price was the pig herd, amounting to 34.85% impact after 18 months. The importation of pork had a small structural impact with the fluctuation of price, within the first four months had a positive impact and subsequent negative impact. The main factor that caused the fluctuations of pork price by changing the market relation between the demand and supply was the pig epidemic, with the current month’s epidemic index having a 9.78% impact on pork price after 18 months. The researcher’s recommendation based on the results of the study is to focus on monitoring and early warning of the

pork price by analyzing big data, strengthening the implementation of control measures of early prevention during disease outbreaks to stabilize pork market supply and price.

According to Gomez, C. J. J. (2025), pork prices have risen sharply, impacting consumers and farmers. This article explores price trends, market outlook, and government initiatives, including DOST-PCAARRD projects, to stabilize supply and strengthen the swine industry. The Department of Agriculture (DA), through its price monitoring database Bantay Presyo, reported January 24, 2025, a surge in pork prices across public markets in Metro Manila. The average retail price per kilogram (kg) of pork reached Php400, which is an increase of 11.11% compared to the previous month's average price. The price in Metro Manila was also evident in other areas. As of January 28, 2025, the average retail price of pork at the Batong Malake Public Market in Los, Baños, Laguna, rose from last week's average of Php400 per kg to Php440 per kg. This indicates that pork retail prices continue to climb.

According to the Philippine Statistics Authority (PSA), prices of essential goods such as rice, pork, and vegetables fluctuate regularly, affecting both producers and buyers (Philippine Statistics Authority, 2025). Monitoring the prices of agricultural goods and livestock is important for farmers, traders, and consumers. Prices can change due to factors like supply, demand, weather conditions, and government policies. Studies have shown that tracking these changes helps in making better business decisions and prevents losses. According to the Philippine Statistics Authority (PSA), prices of essential goods such as rice, pork, and vegetables fluctuate regularly, affecting both producers

and buyers (Philippine Statistics Authority, 2025). By using analytics, price trends can be studied to understand market movements and make predictions for future prices.

Price monitoring plays a crucial role in ensuring fair market conditions for agricultural goods and livestock. In the Philippine rice market, price fluctuations are influenced by factors such as production declines, import surges, and government policies (Gomez, 2024). The Department of Agriculture (DA) monitors price trends and implements interventions to stabilize costs, such as inspecting markets and enforcing price regulations. Despite efforts like tariff reductions and increased imports, retail rice prices remain high, raising concerns about market manipulation. Predictive analytics can enhance price monitoring by analyzing historical trends and forecasting future price changes, helping farmers, traders, and policymakers make informed decisions. By integrating data-driven solutions, stakeholders can improve supply chain efficiency and ensure stable pricing in the agricultural sector.

Price monitoring is essential to stabilizing the cost of agricultural goods and livestock, ensuring that consumers and producers are protected from price manipulation. The Department of Trade and Industry (DTI) and the Department of Agriculture (DA) have strengthened their collaboration with local government units (LGUs) to enforce price monitoring regulations and maintain fair pricing in wet markets (DTI, 2021). Through initiatives like the Suggested Retail Price (SRP) policy and expanded market monitoring, these agencies aim to prevent unreasonable price spikes caused by supply disruptions and external factors. The use of data analytics and predictive tools can further improve price

monitoring by identifying trends and forecasting price changes, enabling better decision-making for stakeholders in the agricultural sector. Effective price regulation and predictive analytics can help stabilize market conditions, ensuring fair trade and food security.

Domingo, S. N. Et al. (2022) provide a comprehensive domestic benchmarking analysis of the Philippine livestock, dairy, and poultry industries, highlighting that these private sector-led sectors contribute significantly to the country's agricultural output despite historically limited government support. Their study details how challenges such as the African Swine Fever outbreak in 2019 and the COVID-19 pandemic in 2020 have negatively impacted production volumes and market dynamics, particularly in the swine and poultry sectors, while dairy production has continued to grow—albeit with most locally consumed milk being imported. The report emphasizes that most domestic production is still carried out by backyard operations, suggesting that consolidating these smaller producers under farm organizations could harness economies of scale, improve capacity, and boost competitiveness against imported products, ultimately strengthening food security and resilience against future shocks.

Briones and Espineli (2022) conducted a consolidated benchmarking study comparing the performance of the Philippine livestock, poultry, and dairy industries with those in major Asian countries such as China, Thailand, and Vietnam. Their work highlights how economies of scale play a significant role in determining the unit cost of production, noting that commercial operations generally achieve lower costs than backyard farms, despite local production

challenges like high feed costs driven by tariffs on corn imports. The study also points out that while dairy operations at a semi-commercial scale can be profitable, backyard dairy remains less competitive, emphasizing the need for regulatory reform and enhanced technical assistance. These findings underscore the importance of using price monitoring and analytics to assess industry performance and inform policy interventions that can improve competitiveness and overall food security in the sector.

Devendra and Chantalakhana (2021) explore how efficient natural resource management in animal production can play a crucial role in alleviating rural poverty and enhancing food security by improving household income and nutritional stability. Their study highlights that, in mixed crop–animal systems, animals contribute significantly to both short-term savings and long-term productivity, especially among poor farming communities. The research emphasizes that improved animal production systems—particularly through targeted investments in technology and collaborative public–private partnerships—can boost the overall performance of the livestock sector. In the context of the Philippines, these findings are particularly relevant as they suggest that integrating advanced price monitoring and predictive analytics with efficient natural resource management can lead to more stable production costs, better market pricing, and ultimately, enhanced competitiveness and food security.

Shively, Barbier, and Burgess (2001) examine the relationships between agricultural intensification and labor allocation at the forest margins in the Philippines. Their study integrates models of lowland agricultural production

and upland labor allocation to show that improvements in labor productivity and higher agricultural wages—spurred by irrigation development—increase labor demand, thereby reducing rates of forest clearing. While the primary focus is on environmental and labor market changes, these findings indirectly relate to price monitoring and prediction. Changes in production practices, labor costs, and land use can alter production costs and market supply dynamics, which in turn affect the pricing of agricultural goods and livestock. Such interconnections are critical for developing analytical models that forecast price changes in response to shifts in production conditions and resource allocation.

Mariano, Villano, and Fleming (2012) examined the socioeconomic, institutional, and environmental factors influencing Filipino rice farmers' adoption of modern technologies and integrated crop management practices. Using binary logit and Poisson models, they found that farmers with higher formal education, access to irrigation, ownership of agricultural machinery, and strong extension services were significantly more likely to adopt certified seed technologies and other best management practices. Conversely, soil deficiencies and risk aversion negatively impacted adoption rates. Their findings underscore the importance of targeted policy interventions—such as improved access to credit, capacity-enhancement activities, and better irrigation facilities—to encourage technology uptake, which in turn can boost rice productivity, enhance food security, and improve the livelihoods of rural communities.

Synthesis and Relevance of the Study

This part summarizes the main insights from the reviewed literature and highlights their significance to the current research. The current body of research emphasizes the critical role of real-time price monitoring and predictive analytics in mitigating agricultural market volatility, driven by factors such as supply-demand imbalances, seasonal trends, and external shocks (Piotr et al., 2015; Supriya & Mamilla, 2022). Studies demonstrate the effectiveness of hybrid models (e.g., ARCH/GARCH, PSO-GA, VMD-SGMD-LSTM) and ICT tools (e.g., mobile-based platforms) in enhancing price transparency and forecasting accuracy (Khabibah et al., 2019; Li et al., 2024; Zhang & Tang, 2024). These studies together highlight the transformative potential of real-time price monitoring and market prediction systems, particularly for hyperlocal contexts like public markets, which aligns with the goals of this research. However, gaps persist in deploying accessible, Android-based solutions tailored to small-scale traders and consumers—a gap this study addresses by developing a data-driven application for Los Baños Public Market.

Numerous researchers have explored real-time price monitoring and market prediction systems for agricultural commodities, pinpointing elements like supply-demand imbalances, seasonal trends, and external shocks (e.g., weather, epidemics). The results from these studies indicate that advanced models (e.g., ARCH/GARCH, hybrid AI algorithms) and ICT tools (e.g., mobile-based analytics) significantly improve price transparency and forecasting accuracy (Khabibah et al., 2019; Li et al., 2024). Nonetheless, a gap still exists

in the comprehension of hyperlocal, real-time price dynamics in public markets, particularly in integrating Android-based analytics for small-scale traders and consumers. This research intends to tackle by developing a data-driven mobile application tailored to Los Baños Public Market, combining real-time monitoring with predictive modeling to enhance decision-making for stakeholders.

By incorporating these findings, the current research expands on existing knowledge while presenting fresh viewpoints on real-time price monitoring and predictive analytics for agricultural commodities in public markets. The study builds upon established volatility models (ARCH/GARCH) and hybrid forecasting approaches (PSO-GA, VMD-SGMD-LSTM) while introducing innovative mobile-based data collection and analysis specifically tailored for Los Baños Public Market. This integration guarantees that the research is based on current scholarship while offering a significant addition to the discipline through its unique combination of three critical elements: (1) hyperlocal Android-based implementation addressing the gap in grassroots market tools, (2) integration of multi-model forecasting adapted for Philippine agricultural commodities, and (3) real-time data crowdsourcing mechanisms inspired by successful ICT applications in developing markets.

Moreover, the literature emphasizes the need to translate advanced predictive models into accessible, user-centered systems suitable for real-world public markets. This highlights the importance of solutions that balance analytical accuracy with usability and local relevance, ensuring that real-time price information and forecasts effectively support decision-making for small-scale traders and consumers.

Literature Matrix

Table 1. Literature Matrix

Author(s) & Year	Research Focus	Methodology	Key Findings	Relevance to Current Study
Piotr et al. (2015)	Price volatility in Polish agri-commodities	Empirical analysis (govt. data)	Global crises amplify volatility, affecting producer/consumer welfare.	Highlights need for volatility monitoring in public markets.
Khabibah et al. (2019)	Big cayenne price volatility (Indonesia)	ARCH/GARCH models	Low volatility at producer/consumer levels.	Supports use of GARCH for local price analysis in Los Baños.
Calvo (2006)	Sudden stops & price volatility	ARCH techniques	Balance sheet effects drive volatility during economic shocks.	Validates need for real-time monitoring during supply/demand disruptions.
Supriya & Mamilla (2022)	Seasonality in commodity prices	GARCH model	Seasonal supply-demand imbalances complicate price discovery.	Seasonal adjustments crucial for Los Baños market predictions.
Rajon et al. (2014)	ICT for price transparency (Bangladesh)	SMS-based price reporting	Mobile tech reduces price manipulation by middlemen.	Android app can replicate this with enhanced features (e.g., crowdsourced data).
Li et al. (2024)	Scallion price forecasting (China)	PSO-GA hybrid model	Reduced MAPE by 30% vs. traditional models.	Hybrid models (e.g., PSO-GA + LSTM) could improve Los Baños price predictions.

Fafchamps & Aker (2015)	Mobile phones in West African markets	Field experiments	User-generated price data increased farmer profits by 15%.	Supports crowdsourcing via Android app for hyperlocal price accuracy.
Zhang & Tang (2024)	Agricultural futures price prediction	VMD-SGMD-LSTM hybrid	Outperformed traditional models in accuracy (+22%).	Framework adaptable for Philippine commodity futures (e.g., rice, pork).
Pang et al. (2023)	Pork price drivers (China)	SVAR model	Pig epidemics caused 9.78% price fluctuations over 18 months.	Epidemic tracking (e.g., ASF) should be integrated into Los Baños price alerts.
Cayabyab, G. T., Castillo, R. E., Castro, P. J. M., & Ramos, W. V. J. (2019)	Development of a price monitoring mobile app using exponential smoothing for agricultural products.	System development; application testing based on ISO/IEC/IEEE 29119 standards.	Accurate price forecasts; high user satisfaction.	Provides a model for integrating predictive algorithms into mobile tools to support pricing decisions in agriculture.
Kamal & Bablu (2023)	Impact of mobile apps on smallholder farmer empowerment.	Literature review and case studies.	Improved info access and farmer autonomy.	Highlights the transformative role of mobile apps in rural agricultural ecosystems and supports tech-based interventions for farmer empowerment.
Bista et al. (2018	Price and demand forecasting via mobile app.	Forecasting algorithm integration and app testing.	Effective market predictions for farmers.	Validates the role of analytics in reducing risk and improving crop planning.

Conceptual Framework

This conceptual framework describes a predictive price intelligence system designed to serve both administrative and end-user needs through a structured data flow and continuous improvement loop. The system is anchored by a central processing block consisting of Data Pre-Processing, a Forecasting Engine, and the Application Backend. Raw market data, including the Current price submitted by seller, first enters the system, is cleaned and prepared in the pre-processing stage, and is then fed into the Forecasting Engine. This engine is the core intelligence, responsible for running predictive models to generate crucial Predicted Prices and performing in-depth Trend analysis.

The processed information and predictions are then distributed to the two primary user interfaces. The Admin Dashboard acts as the control center, providing administrators with comprehensive tools for managing the system and conducting deep analysis, including access to detailed Data tables, Price history, and the ability to generate PDF reports and monitor Active sellers. Simultaneously, the User Dashboard (mobile) provides end-users with actionable, on-the-go data, focusing on immediate relevance like the Current price of commodities, Predicted Prices, Price differences indicators, and the ability to view data across multi-location and utilize Offline Viewing for enhanced accessibility.

A critical feature of this framework is the integrated Feedback Loop. User feedbacks, collected from interactions and system usage, drive the intake of New daily data. This new data is then used to perform forecast refits, where the Forecasting Engine is retrained or fine-tuned. This continuous loop of data

collection, model recalibration, and re-deployment ensures that the system's accuracy steadily improves, guaranteeing that the predicted prices and analyses remain current, relevant, and highly reliable over time. This makes the entire framework a living, adaptive system rather than a static reporting tool.

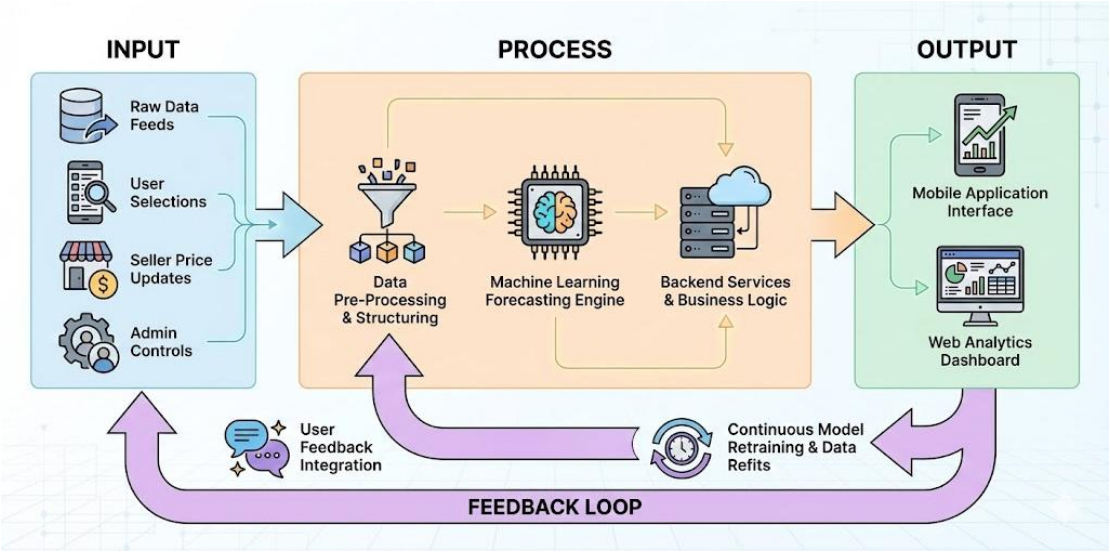


Figure 1. Conceptual Framework

CHAPTER 3

METHODOLOGY

This chapter gave an outline of the research methods that were followed in the study. It provides information on the participants, that is, the criteria for inclusion in the study, who the participants are, and how they are sampled. The researchers described the research design chosen for this study's purpose and the reasons for this choice. The instrument used for data collection is also described, and the procedures followed in carrying out the study are included. The researchers also discuss the methods used to analyze the data that are gathered.

Project Design

The system is a multi-platform solution featuring an Android mobile app and a web-based administrative backend, aimed at delivering real-time price tracking and predictive analytics for agricultural and poultry products in District 2 of Laguna, focusing on the public markets in Cabuyao City, Bay, and Los Baños. This system was designed to enhance transparency, efficiency, and data-informed decision-making for market administrators, vendors, and consumers by incorporating digital tools and smart analytics.

The system architecture illustrates the complete workflow of how market price information is gathered, validated, processed, and delivered to end-users. The process begins with the Market Officer or Administrator, who oversees data collection from different public markets such as Cabuyao, Bay, and Los Baños. These officers are responsible for reviewing and approving the price data

submitted by sellers. Their involvement ensures that only accurate and verified information is integrated into the system.

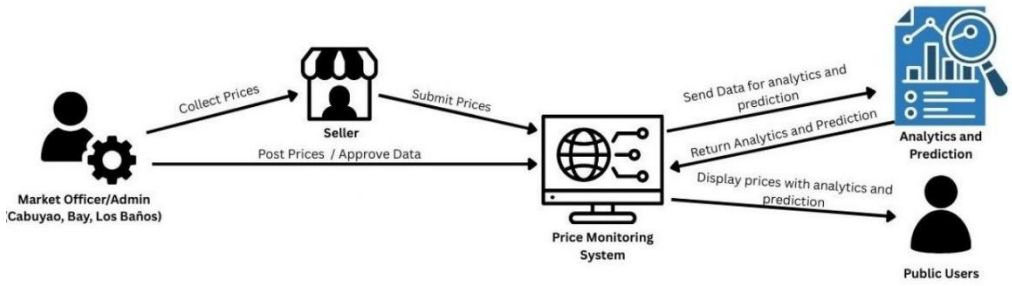


Figure 2. System Architecture

Sellers also play a key role by entering or submitting the prices of their commodities through the system interface. Once sellers submit their data, it is forwarded to the market officer for verification. Only after approval can the price entries be officially posted. This step helps maintain transparency and prevents the publication of incorrect or unverified prices. Both seller input and officer validation work together to create a reliable source of market information.

After approval, the validated data is stored in the Price Monitoring System, which acts as the core platform for managing and organizing the information. The system also connects to a dedicated Analytics and Prediction Module, to which it sends the collected data. This module analyzes historical and current price trends, applies forecasting techniques, and generates insights that can help users understand market behavior. Once the analysis is completed, the module returns the predictive results and analytical summaries to the system.

The Price Monitoring System then displays the updated prices together with the analytical information and predictions. Public Users can access the platform to view real-time price updates, compare trends across dates or commodities, and use predictive insights to guide their purchasing or trading decisions. This architecture supports a streamlined flow of information, ensuring that users receive accurate, timely, and data-driven updates about market prices.

Project Development

The development of the Android-based application and its supporting web system follow the Agile-Scrum methodology, allowing iterative design, regular user feedback, and incremental improvements. The process is divided into the following key phases:

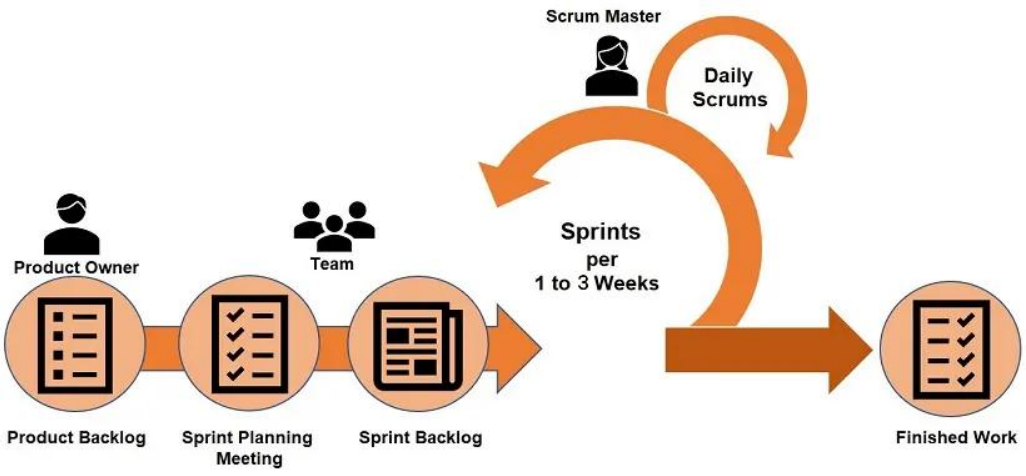


Figure 3. Agile-Scrum Methodology

The design and development of the Android application and its web system for this capstone project follow the Agile-Scrum methodology for iterative development, continuous feedback from stakeholders, and

incremental improvement. As shown in the diagram, Scrum rests its workflow on 2 to 4 weeks of time-boxing sprints during which activities like sprint planning, daily stand-ups, development, sprint review, and retrospectives are performed. Core features like real-time price input, trend visualization, and predictive analytics were defined with critical stakeholders (the vendors, users, and market administrators) at the beginning of the project. These requirements were converted into user stories that were prioritized in the product backlog.

System design and development progressed through multiple sprints. Wireframes and UI prototypes were developed for mobile and web interfaces, while system architecture was set up to cover database schemas, API endpoints, and analytics components. Technologies like Firebase, Python, and Java/Kotlin were chosen for data storage, analytics, and Android development. Development was broken into three major components: the mobile app for end-users, a web-based admin panel for data validation and system management, and the analytics engine running machine-learning models like ARIMA and LSTM. A functional increment of the system was delivered at the end of every sprint, ensuring continuous integration of the feedback and adaptation to new needs that would arise.

Project Testing and Evaluation

To ensure the quality, reliability, and usability of the system, a series of rigorous testing and evaluation procedures are conducted throughout the development lifecycle. These tests focus on verifying system functionality, accuracy of predictions, and user experience across all platforms.

Testing Phases

Unit Testing

The Unit Testing Life Cycle ensured each software component functions correctly before integration. It begins with Requirement Analysis, where testing needs are identified, followed by Test Case Design to create specific test scenarios. Then, the Test Environment Setup prepares the necessary tools and conditions to execute the tests. In Test Case Execution, the tests are run to validate functionality. Any issues found are logged during Defect Reporting and Tracking, after which Re-testing and Regression Testing are performed to confirm fixes and ensure no new bugs were introduced. Finally, Unit Test Closure wraps up the process by documenting results and ensuring all issues are resolved before moving to the next phase.

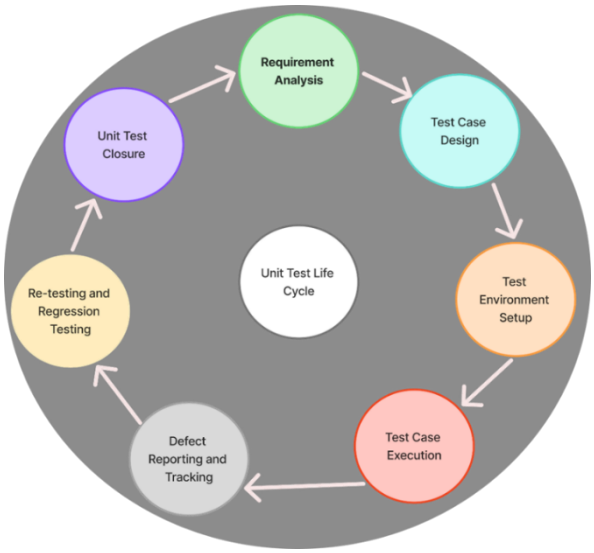


Figure 4. Unit Testing

Integration Testing

The purpose of the integration testing is to verify that all modules within the system interact seamlessly, with particular focus on the connection between the mobile application, the database, and the analytics engine. This testing ensures that data entered by users correctly flows into the database, real-time prices are accurately retrieved and displayed within the app, and the predictive outputs generated by the analytics engine are properly rendered for users. By concentrating on these critical areas, the testing process guarantees data consistency, system reliability, and an overall smooth user experience.

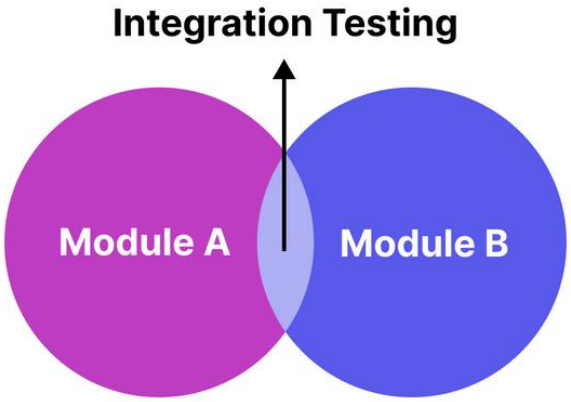


Figure 5. Integration Testing

System Testing

Compatibility Testing. The research was conducted to assess the system's compatibility across a diverse range of devices and web browsers. This comprehensive evaluation involved testing the system's functionality, performance, and user experience on various Android devices and popular browsers such as Firefox, Chrome, Edge and Safari. By conducting rigorous testing across these platforms, the research aimed to identify and address any

compatibility issues, ensuring a seamless and consistent user experience regardless of the device or browser used to access the system.

Load testing. This is where the researchers test the performance of the system to check for performance issues such as system crashes and slow response time and how fast can be processed. And it loads too many tabs in a single browser to test its performance.

Stress testing. Stress testing is where the researchers test the number of users that the website can handle. The researchers do this by accepting multiple numbers of users.

Acceptance Testing

The consideration of this acceptance testing exercise is the project for the most part, to find out if the system satisfies the expectation and requirement of the stakeholders. Users from District 2's Public Market, including vendors, consumers, and market administrators, interact with the system in real scenarios performed by the application. In task-based testing, participants would perform specific actions while developers observe the system performance and capture the feedback using structured forms, thereby validating the system's usability, functionality, and overall satisfaction prior to any complete deployment.

Evaluation Procedure

On this premise, the system would be evaluated concerning functionality, performance, usability, and prediction accuracy, allowing real users from Los Baños Public Market to participate in the process.

Functional Testing. ensure that each module, including the mobile app, admin panel, and prediction model, serves the purpose for which it was designed. The testing a combination of manual and automated test cases focusing on critical features like price submission, user login, admin validation, and price forecast display.

Usability Testing. was aimed to measure how easy the system is to be used for the farmers, vendors, and administrators. The methodology for this involved a structured interview, feedback sessions with users, and then administering a System Usability Scale (SUS) survey for the quantitative measure of user satisfaction.

The Evaluation of Prediction Accuracy. is an evaluation on the prediction algorithm's reliability. Using historical price data, predicted values are compared to actual results. The key metrics used in measuring the accuracy of the model include Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE).

Performance Testing. is intended to evaluate the speed and reliability of real-time data updates and the general responsiveness of the system. This requires simulating a sufficient number of concurrent users and testing load times, latencies, synchronization performance of updates, etc.

Population of the Study

The population for this study includes key stakeholders involved in the agricultural and poultry trading activities across District 2 of Laguna, covering

the public markets in Cabuyao City, Bay, and Los Baños. The primary participants are:

Vendors/Sellers – Stall owners and market vendors responsible for setting prices and selling goods.

Consumers – Regular buyers, including households and small business owners who rely on market products.

Market Administrators – Officials and personnel responsible for overseeing market operations, pricing regulations, and data management.

A purposive sampling technique was be used to select approximately 15 to 60 respondents, ensuring balanced representation from each stakeholder group. This approach facilitates comprehensive data collection and user feedback for the system's development and evaluation.

Project Locale

The project cover District 2 that includes Cabuyao City, and the municipalities of Bay and Los Baños, Laguna, Philippines, specifically targeting public market areas where agricultural and poultry products are actively traded. It was chosen for the highly dynamic nature of its marketplace. Common features include high transactions among buyers and sellers, fluctuating prices of commodities, and an increasing presence of digital technology within the market environment. The location would be suitable for a real-life scenario for a pilot rollout and an iterative testing ground for the application on Android and its web-based support system.

The data to be collected from this area also provide an input to the machine-learning algorithms of the system in developing accurate price forecasting models that are contextualized. This consideration of District 2 ensures that the project to be much more relevant to the local economic behavior, seasonal trends in pricing, and decision-making processes of the community. The geographical focus also vibrates with the larger aspirations of promoting digital inclusion and market transparency in local agricultural economies.

Product Backlog

The product backlog serves as the foundational guide for the system's development, outlining critical features that align with user and administrative needs. Each backlog item is structured as a user story to highlight its purpose and intended outcome. The system prioritizes high-importance functionalities such as user registration and login, seller authorization, price submission, and price trend prediction, which collectively ensure seamless interaction and transparency between buyers and sellers. Administrative tasks, including user and product management, data verification, and approval or rejection of seller accounts, are also emphasized to uphold system accuracy, security, and operational integrity. This backlog ensures that development efforts remain user-centered while maintaining robust backend support and control. User registration, seller authorization, and price submission ensure secure and reliable data, while price trend prediction provides market insights. Administrative features support access control and data accuracy.

Table 2. Product Backlog

Code	Backlog Item	User Story	Priority	Description
PB01	User Registration/Login	As a seller or buyer, I want to create or log into my account so I can use the app.	High	Handles both new account creation and returning user login.
PB02	Seller Authorization	As a seller, I want my account to be verified before I can submit product prices.	High	Admin verifies seller data before allowing price submissions.
PB03	Price Submission	As a verified seller, I want to input product prices so buyers can see market updates.	High	Sellers can submit product prices after successful login and authorization.
PB04	Price Trend Prediction	As a user, I want to see price trends to help predict future prices.	High	The analytics system displays predicted price movements.
PB05	Admin: User Management	As an admin, I want to manage users so I can control access and permissions.	High	Admin handles registration approvals and data verification.
PB06	Admin: Product Management	As an admin, I want to manage product data to ensure accuracy in displayed prices.	High	Admin monitors and manages product listings.
PB07	Admin: Data Verification	As an admin, I want to verify user-submitted data to maintain system integrity.	High	Admin checks the accuracy and validity of submissions.
PB08	Admin Approval/Reject Function	As an admin, I want to approve or reject seller applications.	High	Enables acceptance or denial of seller requests.

Sprint Planning

Sprint planning is a crucial ceremony in agile development, offering framework and modus operandi to development. It is to prioritize whatever may be goals, clearly identify what needs to be done, and steer all efforts toward the successful delivery of some features within a specific limited time. The whole project is hence broken down into small sprints so that some heavier work can be going with the trend of changing requirements and challenges.

This method bridges the gap between intending and focuses definition, making sure every member is aware of what would be done, by whom, when, and why. This is also a way of increasing collaboration among many, therefore optimizing the use of resources and creating a more focused and efficient process for development. Each sprint always ends with a review of the work done and thus becomes a source of continual improvement in relation to the project goal.

Table 3. Sprint Planning

Sprint	Duration	Sprint Goal
Sprint 1	Weeks 1 to 2	Build the basic structure of the app, user login, and admin login.
Sprint 2	Weeks 3 to 4	Add the main features like price updates, admin tools, and charts.
Sprint 3	Weeks 5 to 6	Add prediction models, maps, testing, and prepare for final defense.

System Flowchart

This system is structured around two distinct workflows, one for users via the Mobile Application and one for administrators through the Web-based System, to manage price information with control and reliability. In the user flow, the mobile application first identifies the user's role; Buyers are granted view-only access to essential market data, including the Commodity Price List and Price Forecast, and can filter this information by changing locations; conversely, Sellers must pass authentication to access the Price Input Module to submit the prices for later administrative verification, with all verified seller inputs being aggregated to compute an average price used as an administrator reference.

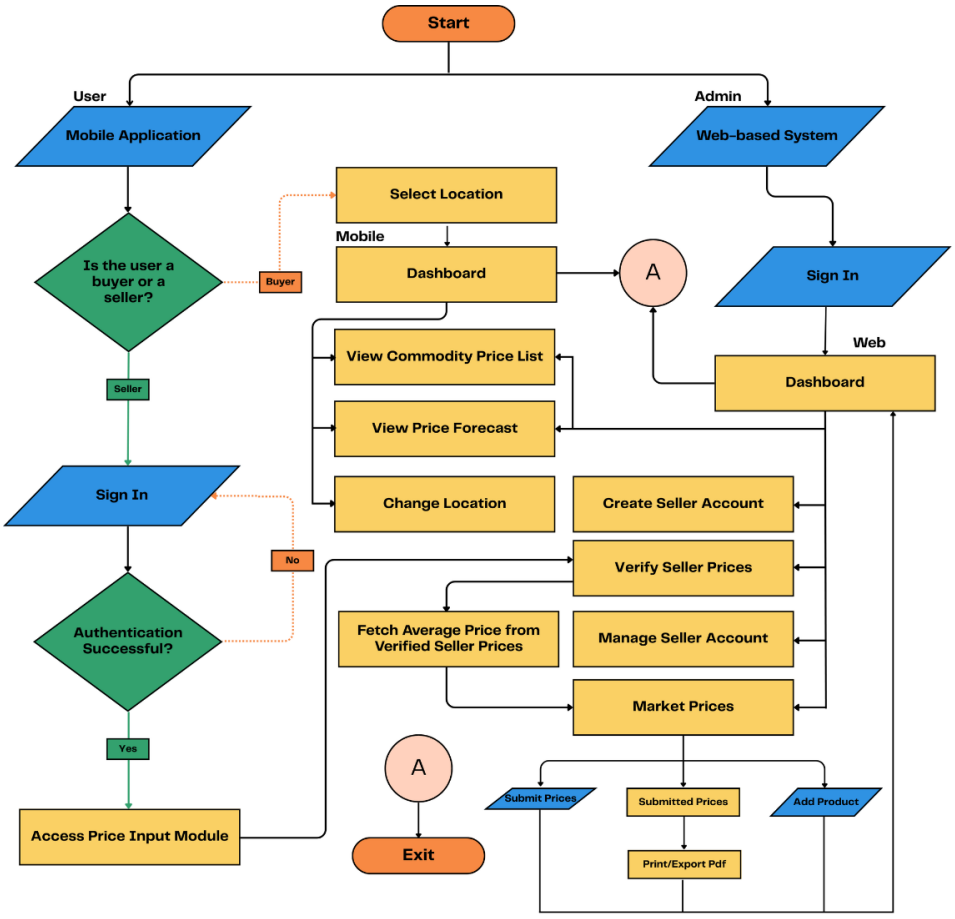


Figure 6. System Flowchart

The administrator workflow requires secure authentication to access management modules where the administrators can create and manage seller accounts, verify seller-submitted prices, crucially, and manually enter the final, official prices for each commodity, a decision guided by the calculated average of verified seller submissions, thereby ensuring the admin-submitted prices are informed by market trends yet reflect expert judgment and form the basis of the reliable price list and forecast presented to end-users; administrators can also track price history, manage accounts, export reports, and add new products, ultimately maintaining the integrity and accuracy of the system's market data across both platforms through structured, role-based interaction..

Use Case Diagram

Figure 7 shows the commodity data system involves several key Actors and their associated Core Use Cases. The primary consumers are the User (general public) and the Seller, who interact via a Mobile/Web App. Both engage in General Viewing, which involves selecting criteria (commodity, location, date), viewing current or historical data, and crucially, viewing Price Forecasts, which depend on the system's backend prediction engine. The Seller also serves a vital role in Data Contribution by submitting real-time local prices, enriching the dataset. Maintaining system integrity falls to the Admin via a Web Portal, whose System Management responsibilities include verifying submitted prices, managing user accounts, and managing the range of tracked commodities. The system also has an essential non-human actor, the Timer/Scheduler, which is responsible for Backend Automation.

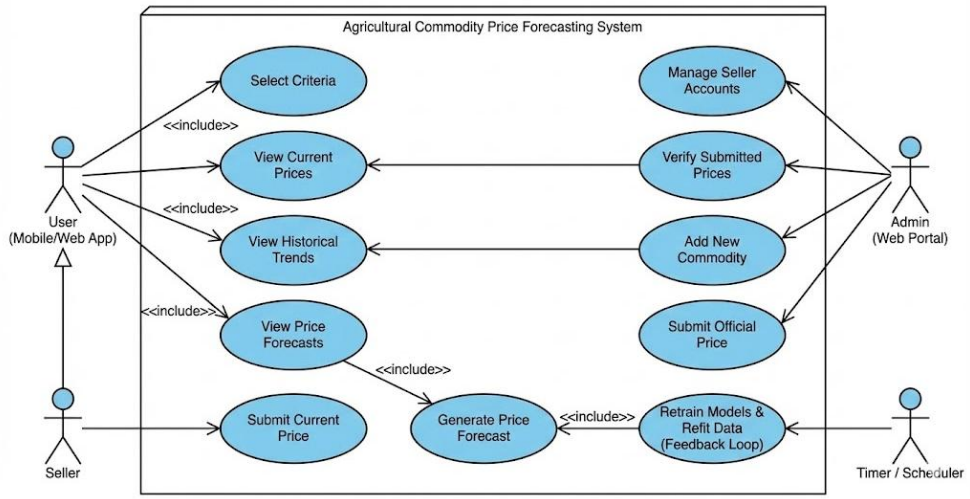


Figure 7. Use Case Diagram

This automation includes the system's core function, Generate Price Forecast, and the critical Feedback Loop process: Retrain Models & Refit Data. This retraining step, often triggered every 1 and 15th of the month by the Timer, ensures the ML models remain accurate by incorporating the newest verified data.

Materials

Software Tools

Android Studio. is used to develop the mobile application that would run on Android devices, utilizing Java or XML programming languages. It provides a powerful, integrated development environment (IDE) with built-in tools for designing user interfaces, testing, and debugging mobile apps efficiently.

Python. is utilized for data analytics and price forecasting, employing Pandas and NumPy for data manipulation and preprocessing, StatsForecast for time series modeling using the Seasonal Naive approach, and Scikit-learn

for performance evaluation through MAE and MAPE metrics. SQLAlchemy facilitates database connectivity, ensuring accurate, efficient, and data-driven market trend forecasting.

AWS or **phpMyAdmin.** Serve as the system's real-time database solution. AWS (Amazon Web Services) offers scalable cloud-based storage and real-time data management ideal for mobile and web applications, while phpMyAdmin provides an efficient web-based interface for managing and organizing structured databases, allowing easy handling of pricing and user data.

GitHub or GitLab. Serve as the version control and team collaboration platforms. These tools allow developers to track code changes, manage development branches, collaborate in real-time, and maintain the integrity of the project across different stages of development.

Hardware

Table 4 show the minimum requirement for specification of the computer used to develop the system. The working computer is running on Windows 10 above with 8gb installed RAM and using Intel Core i5.

Table 4. Hardware Specification for PC/Laptop

Minimum Requirement	Specifications
RAM	8GB
CPU	Intel Core i5
Operating System	At least Windows 10 above

Table 5 show the minimum requirement for specification of the android phone used to develop the system. The working android phone is running on android 10 above with 8gb installed RAM and using Quad-core processor.

Table 5. Android Phone Specification

Minimum Requirement	Specifications
Android OS	at least Android 10 above
RAM	8GB
Processor	Quad-core

Sprint Backlog

Sprint Backlog provides a clear approach for developing the core functionalities of our application in four main sprints, each of which focuses on a specific priority task from the product backlog for incremental and smooth delivery of the system.

The Sprint Backlog outlines a three-phase development plan aimed at delivering a fully functional mobile application that supports both users and administrators in managing and accessing real-time product pricing. Sprint 1 focuses on the foundational setup of the system, including the creation of the Android app structure, database integration, and user interface for login and registration. This sprint ensures that both users and admins can securely access the system. Sprint 2 transitions into the core functionality, where sellers can input prices, and users can view real-time updates. An admin dashboard is developed for managing data submissions, along with product filtering features

and visual representations of past price trends. Integration testing is conducted to confirm system coherence.

Each sprint has its own set of tasks that need to be finished during that two-week period. Below is a simplified list of tasks for each sprint based on the goals:

Table 6. Sprint Backlogs

Priority	Item	Task	Estimated Work Hours
Sprint 1	System Setup and Login	- Set up the Android app structure and connect it to the database. - Design the user interface and login screens. - Add user registration and login features. - Create login access for admins. - Test if the login and registration features work properly.	48–96 Hours
Sprint 2	Core Features and Admin Tools	- Add the form for sellers to input prices. - Show real-time prices in the app. - Build the admin dashboard to review and approve price entries. - Add features to manage products and filter by category. - Include simple charts to show past price trends. - Do integration testing to ensure all features work together.	48–96 Hours
Sprint 3	Analytics and Finalization	- Set up forecasting tools using machine learning to predict prices. - Show predicted prices inside the app. - Add push notifications to alert users about price changes. - Add maps to show vendor locations using GIS. - Test the full system and gather feedback from users. - Prepare final documents and system presentations.	48–96 Hours

Sprint 3 emphasizes intelligent analytics and system finalization. This includes implementing machine learning models for price forecasting, adding

push notifications for market updates, integrating GIS maps to display vendor locations, and conducting full system testing. The sprint concludes with the preparation of final documentation and a system presentation, ensuring a complete and user-friendly platform.

Data Collection Instrument

To gather relevant data for the development and evaluation of the “Price Pulse App” a web-based and mobile-based app, the researchers utilized a survey questionnaire as the primary data collection instrument. The questionnaire was designed to assess the current price checking and tracking for the residents, consumers, and stall owners in the public market of Cabuyao City, and the municipalities of Los Baños and Bay. It also gathered feedback on system usability and key features after prototype testing. The survey included a combination of multiple-choice and Likert-scale allowing qualitative data to be collected. This instrument was distributed to target respondents across District 2, which consists of the following municipalities; Cabuyao City, Los Baños and Bay. The data gathered played a vital role in understanding user expectations and validating the effectiveness of the proposed system in enhancing the market price checking and tracking.

Statistical Treatment

The collected data from the target respondent pool underwent analysis using a combination of population and purposive sampling techniques. This approach enables a comprehensive evaluation of the system's functionality and overall performance during the development process. By employing both 60 population and purposive sampling, a representative sample from the target

respondent pool is obtained, allowing for a thorough examination of diverse perspectives. This method is expected to facilitate the testing and evaluation of the system's functionality and performance. These insights play a crucial role in informing a more tailored and user-centric development approach. The developers also utilized Likert Scale methods to evaluate the application features. The evaluation consisted of a series of questions that may be answered by one of four responses/impressions, which were sorted in the following order: five (5) – Very High four (4) – High, three (3) – Moderate, two (2) – Low, and one (1) – Very Low.

Table 7 outlines the scaling technique planned for the upcoming survey. A Likert Scale is employed to gauge user satisfaction with the application. Developers then apply a percentage-based method to evaluate the respondents' feedback, facilitating a detailed analysis based on the gathered results.

Table 7. Likert Scale Method

RATING	MEAN RANGE	INTERPRETATION
5	4.01 – 5.00	Very High
4	3.25 - 4.0	High
3	2.50 - 3.24	Moderate
2	1.75 - 2.49	Low
1	1.0 - 1.74	Very Low

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results and the discussion to support the methodology utilize in the achievement of the objectives.

The results of the study indicate that the developed system effectively enhances price monitoring, data management, and analytical capabilities for agricultural commodities. User testing showed high satisfaction with its accuracy, usability, and reliability, noting that automated data processing reduces manual workload and minimizes common errors found in traditional methods. Integrated analytics, including trend visualization and predictive insights, were also recognized for supporting clearer and more informed decision-making. Evaluation using ISO 25010 further confirmed strong performance in functionality, efficiency, compatibility, and user experience, demonstrating that the system is technically sound, practical, and beneficial for stakeholders involved in commodity price monitoring.

Administrator Interface

The admin dashboard shows the data from Los Baños that presents real-time market information in a structured and admin-friendly layout, allowing admin to quickly understand commodity trends and price movements.

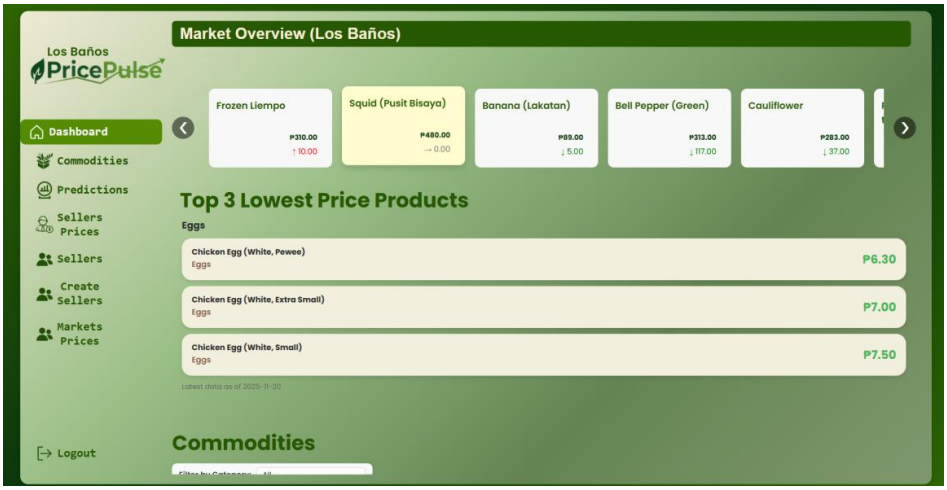


Figure 8. Dashboard Market Overview for Administrator

The commodities analytics page allows the administrators to see a clear overview of the price trends and current information for various agricultural products. The page also shows a line graph of daily agricultural and poultry products, filters for sorting products, and a table with current prices and changes to help administrators monitor commodity prices and maintain accurate market tracking.

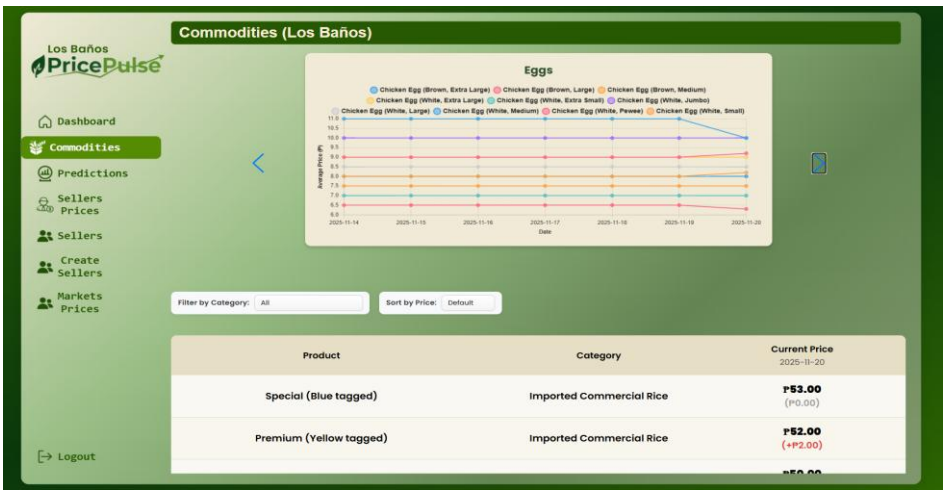


Figure 9. Analytics of Commodities for Administrators

In this section shows the predicted prices of different commodities. At the top, admin can search specific types of commodities and can also filter the categories to lessen search time. Admin can also choose a date to see the price forecast. The graphs help the admin to quickly see if the price is going up, down, or staying steady.

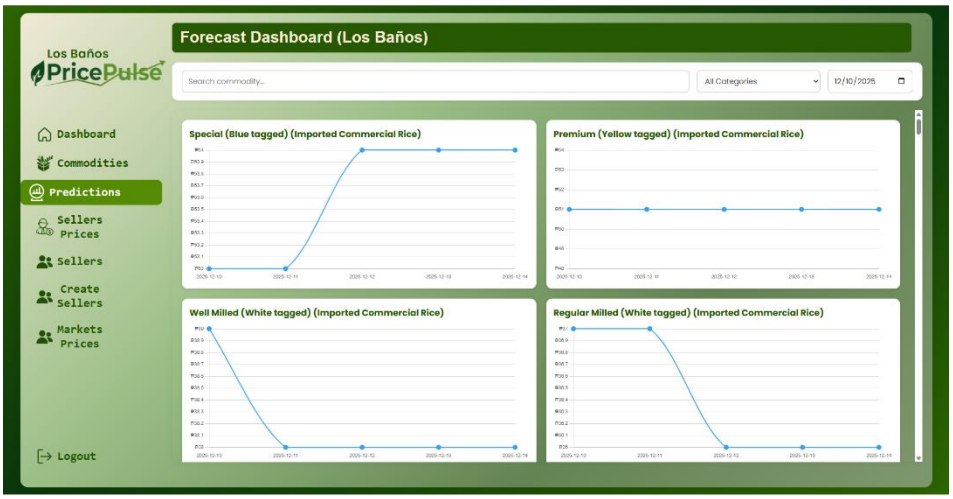


Figure 10. Prediction Page for Administrators

In this section allows the admin to manage and verify all the sellers in the market. The admin can see the seller's product and have the authority to verify and deny the product that is inputted by the seller.

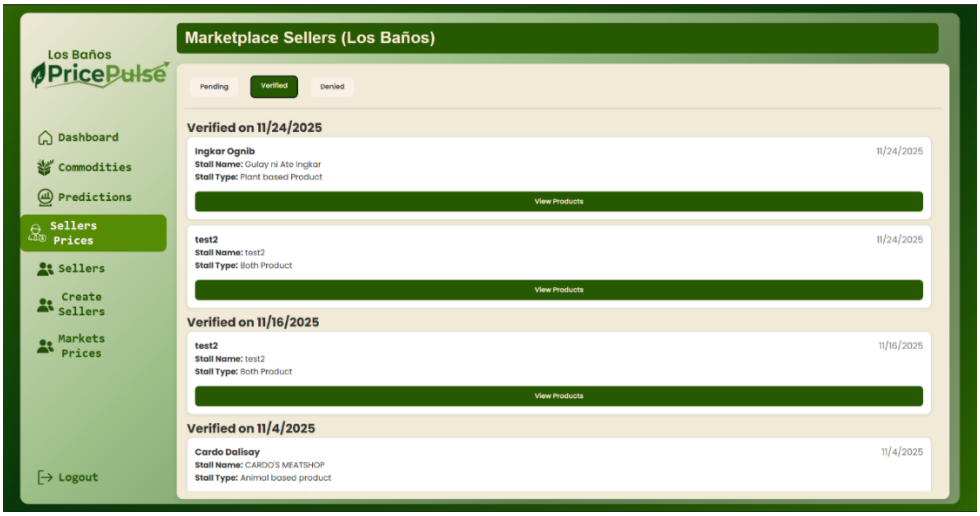


Figure 11. Seller Page for Verifying Products for Administrators

This section is where the admin can see all the stall owners in the market. The admin can also deactivate the stall owner's account if the seller is not active.

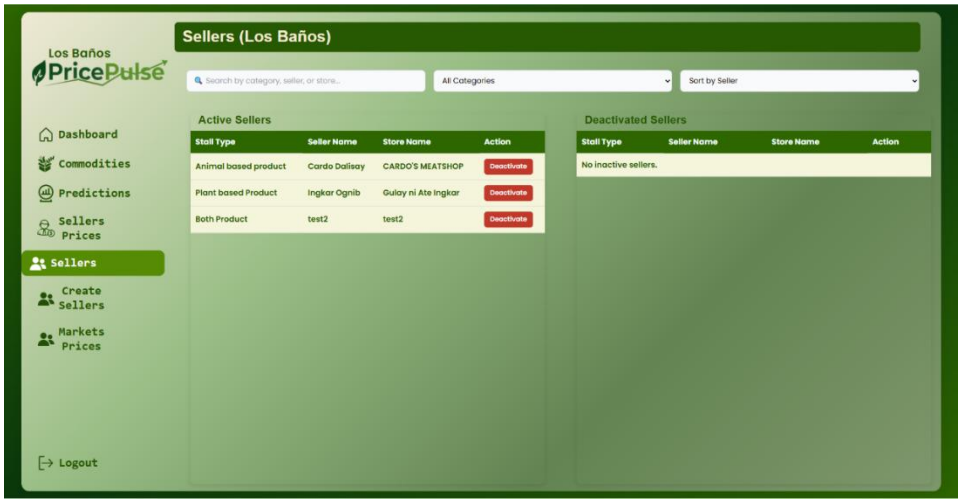


Figure 12. Seller Management for Administrators

This section is where the administrator creates a new seller information to be added to the system. The administrators need to fill-up the required form to create a new seller.

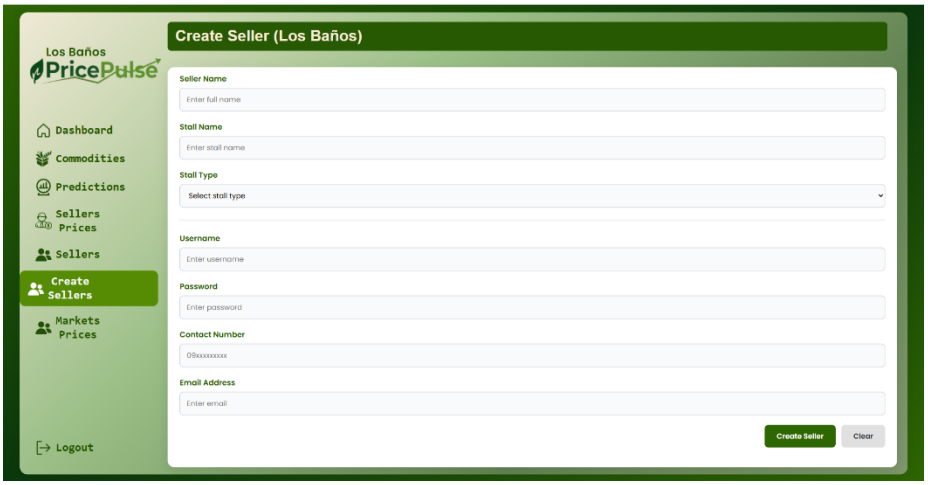


Figure 13. Create Seller Page for Administrators

The Market Prices section allows the administrator input the regular price and verify the submitted prices of the sellers. The administrators can also add a product or commodity if it wasn't on the system.

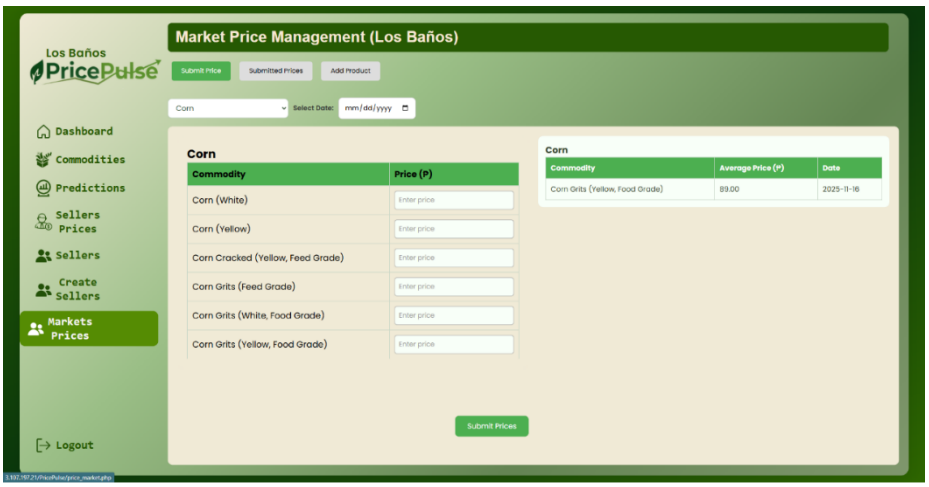


Figure 14. Market Price Management for Administrators

User Interface

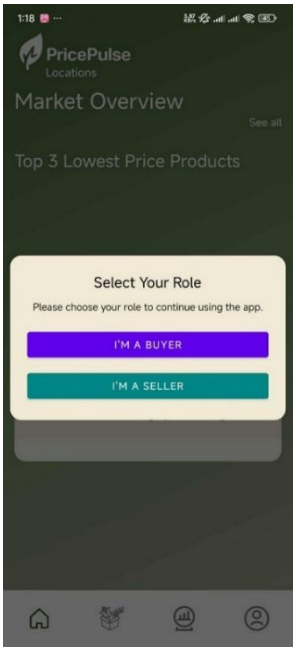


Figure 15. User Role Selection Interface

In this section, users can choose how they will use the application upon entry, either as a buyer or as a seller. Sellers are required to log in to their respective accounts to access features related to managing products, prices, and other seller-specific functions. On the other hand, buyers or general users are not required to create or log in to an account, allowing them to freely view and check regular market prices for reference and comparison. This approach ensures accessibility for users who only need price information while maintaining secure access for sellers who actively use the system.



Figure 16. Seller Interface for Inputting Commodity Prices

This section allows sellers to enter and update the prices of the commodities they offer in the market. Once the prices are submitted, they undergo a review process by the administrator to ensure that they comply with established market pricing standards. Only prices that meet these standards

are approved and reflected in the system, helping maintain accuracy, fairness, and reliability of the market price information presented to users.



Figure 17. Market Overview Displaying Real-Time Commodity Prices

This section displays a comprehensive list of commodities available in the selected location, along with their corresponding categories and current prices. It allows users to search for specific products and filter items by category for easier navigation. Updated price values and recent price changes are clearly shown, enabling users to compare prices across different commodities. This organized presentation helps users monitor market trends and make informed decisions based on accurate and up-to-date pricing information.



Figure 18. Product List Interface

In this section displays a searchable and filterable list of commodities, allowing users to view products by category and check their current market prices.



Figure 19. Predicted Price Interface

This section displays the predicted commodity prices based on the selected date. It also includes a search and category filters, allowing users to refine the list of the products.

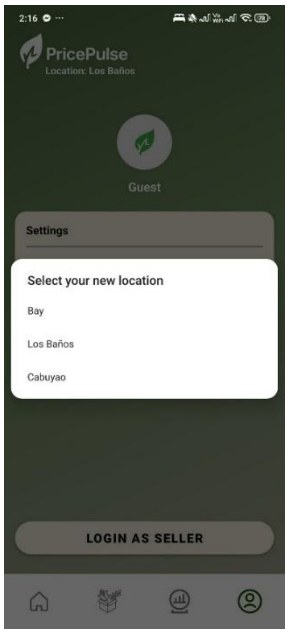


Figure 20. Location Selection Interface

In this section users can select and change the location anytime. Users can view other municipalities market prices.

Evaluation results using ISO 25010 software product standards to verify its efficacy and quality.

Evaluate the system's performance in terms of accuracy, usability, and reliability through user testing, feedback collection, and comparative analysis with existing price monitoring methods using ISO testing.

The system was rigorously evaluated using the ISO 25010 software quality framework, and results showed consistently “Very High” ratings across

all measured quality attributes. Functional Sustainability recorded the highest score at 4.88 ±0.33, reflecting the system's strong ability to perform intended functions accurately and consistently. High ratings for Security and Portability (both 4.80 ±0.41) demonstrate the system's capability to protect user data and operate smoothly across different platforms. In addition, the system exhibited strong Usability (4.70 ±0.46) and Compatibility (4.66 ±0.51), indicating that users found the interface intuitive and the system well-suited for integration with various devices and environments.

Table 8 reveals a consistently positive user perception of the Price Pulse app, with an overall composite mean of 4.42, indicating strong agreement. The result shows that the app helps users perform market price tracking efficiently and accurately, as evidenced by a mean rating of 4.40. The respondents also perceived the app as valuable in providing useful data insights for market-related decision-making, which received one of the highest ratings at 4.53. Moreover, the app was viewed as contributing to improved work productivity, with an average of 4.35. The accuracy and reliability of the information generated by the system were also rated very high in two separate items, with average scores of 4.38 and 4.55, indicating strong confidence in the system's data quality. Overall, the composite mean of 4.42 with a standard deviation of ±0.63 demonstrates that respondents consistently perceived the app as highly useful, dependable, and beneficial for market monitoring and decision-making tasks.

Table 8. ISO 25010 Testing

ITEMS	MEAN	SD	INTERPRETATION
Functional Sustainability	4.88	±0.33	Very High
Performance Efficiency	4.33	±0.50	Very High
Compatibility	4.66	±0.51	Very High
Usability	4.70	±0.46	Very High
Readability	4.50	±0.51	Very High
Security	4.80	±0.41	Very High
Maintainability	4.41	±0.51	Very High
Portability	4.80	±0.41	Very High
COMPOSITE MEAN	**4.63**	±0.45	**Very High**

n=3, Legend: highest to lowest as follows: five (5) – Very High (4.21 – 5.00), four (4) – High (3.41 – 4.20), three (3) – Moderate (2.61 – 3.40), two (2) – Low (1.81 – 2.60), one (1) – Very Low (1.00 – 1.80).

Other evaluated aspects—including Performance Efficiency (4.33 ±0.50), Readability (4.50 ±0.51), and Maintainability (4.41 ±0.51)—also received “Very High” interpretations, affirming that the system is fast, easy to understand, and simple to update or enhance. With an overall composite mean of 4.63 ±0.45, the findings confirm that the system meets high-quality software standards and performs reliably, making it well-suited for real-time agricultural and poultry price monitoring and long-term operational use.

Evaluation results using TAM software product standards to verify its efficacy and quality.

Table 9 shows that respondents generally found the price pulse app very easy to use, with most indicators receiving very high ratings. The first statement received an average of 4.38, indicating that users believe the system can be understood easily. The interface was rated very high by the users, with a mean of 4.53. The users agreed that the system's features are easy to locate and operate, reflected in a mean rating of 4.40. The system's ability to help users recover from mistakes also received a very high rating of 4.35, showing that users feel supported when errors occur. The only item rated high was the system's responsiveness to user actions, with a mean of 4.13 and a standard deviation of ±0.98. Overall, the composite mean of 4.36 with a standard deviation of ±0.80 indicates that the system is perceived to be very easy to use.

Table 9. Perceived of Usefulness

Perceived Usefulness	Mean	SD	Interpretation
The system helps users perform market price tracking tasks efficiently and accurately.	4.4	±0.64	Very High
The system provides useful data insights for market decisions.	4.53	±0.5	Very High
The system contributes to improve work productivity.	4.35	±0.68	Very High
The system provides accurate and reliable information.	4.38	±0.69	Very High
The system provides accurate and reliable information.	4.55	±0.62	Very High
COMPOSITE MEAN	**4.42**	**±0.63**	**Very High**

n=60, Legend: highest to lowest as follows: five (5) – Very High (4.21 – 5.00), four (4) – High (3.41 – 4.20), three (3) – Moderate (2.61 – 3.40), two (2) – Low (1.81 – 2.60), one (1) – Very Low (1.00 – 1.80).

The table 10 reveals a consistent attitude of the user's perception and feelings toward interacting with the system. The results indicate that users generally have a very positive attitude. The users feel confident, with an average of 4.40 using the system and are satisfied with its overall performance, with an average of 4.31. The users also trust the system's reliability and feel in control while navigating the system, that get an average of 4.26. Moreover, the users find the system pleasant and engaging, with an average of 4.40. Overall, the composite mean of 4.32 reflects a very high positive attitude, suggesting that the system is user-friendly, dependable, and provides and enjoyable experience for the users.

Table 10. Perceived Ease of Use

Perceived Ease of Use	Mean	SD	Interpretation
Learning to use the system is easy and intuitive.	4.38	±0.80	Very High
The interface is clear and easy to navigate.	4.53	±0.76	Very High
Features are easy to locate and operate.	4.4	±0.69	Very High
The system helps users recover from mistakes easily.	4.35	±0.75	Very High
The system responds quickly to user actions.	4.13	±0.98	High
COMPOSITE MEAN	**4.36**	**±0.80**	**Very High**

n=60, Legend: highest to lowest as follows: five (5) – Very High (4.21 – 5.00), four (4) – High (3.41 – 4.20), three (3) – Moderate (2.61 – 3.40), two (2) – Low (1.81 – 2.60), one (1) – Very Low (1.00 – 1.80).

The table 11 shows that the respondents strongly intend to continue using the system. The data report show that the users intend to keep using the system, with an average of 4.46. Also, the respondents expressed a very high

willingness to recommend it to others, with a mean of 4.55, suggesting that they perceived it as reliable, effective, and valuable enough to endorse. The users prefer the current system over similar platforms, with an average score of 4.30. The respondents also indicated readiness to integrate the system into daily workflow with a mean of 4.23. The respondents want to adopt an improved version or updates, with a mean of 4.56. Overall, the composite mean of 4.42 reflects a very high behavioral intention to use the system.

Table 11. Attitude Toward Using

Attitude Toward Using	Mean	SD	Interpretation
Users feel confident using the system.	4.4	±0.71	Very High
Users are satisfied with the system's overall performance.	4.31	±0.81	Very High
Users trust the system to function reliably.	4.21	±0.73	Very High
Users feel in control while using the system.	4.26	±0.82	Very High
Users find the system pleasant and engaging.	4.4	±0.61	Very High
COMPOSITE MEAN	**4.32**	**±0.74**	**Very High**

n=60, Legend: highest to lowest as follows: five (5) – Very High (4.21 – 5.00), four (4) – High (3.41 – 4.20), three (3) – Moderate (2.61 – 3.40), two (2) – Low (1.81 – 2.60), one (1) – Very Low (1.00 – 1.80).

Table 12 shows that the respondents strongly intend to continue using the system. The data indicate that users plan to keep using the system, with a mean score of 4.46. The respondents also expressed a very high willingness to recommend the system to others, with a mean of 4.55, suggesting that they perceive it as reliable, effective, and valuable enough to endorse. Moreover, users prefer the system over other similar platforms, as reflected by a mean score of 4.30. The respondents further indicated a strong readiness to integrate

the system into their daily workflow, with a mean of 4.23. Additionally, respondents showed a very high intention to adopt improved versions or future updates of the system, obtaining the highest mean of 4.56. Overall, the composite mean of 4.42 reflects a very high behavioral intention to use the system.

Table 12. Behavioral Intention to Use

Behavioral Intention to Use	Mean	SD	Interpretation
Intends to continue using the system.	4.46	±0.67	Very High
Willingness to recommend the system to others.	4.55	±0.69	Very High
Prefers this system over other similar platforms.	4.3	±0.80	Very High
Willing to integrate this tool into daily workflow.	4.23	±0.78	Very High
Intends to use improved version or updates.	4.56	±0.59	Very High
COMPOSITE MEAN	**4.42**	**±0.71**	**Very High**

n=60, Legend: highest to lowest as follows: five (5) – Very High (4.21 – 5.00), four (4) – High (3.41 – 4.20), three (3) – Moderate (2.61 – 3.40), two (2) – Low (1.81 – 2.60), one (1) – Very Low (1.00 – 1.80).

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATION

Summary

The study of Real-Time Price Monitoring and Market Prediction for Agricultural Commodities in District 2: A Multi-Platform Data Analytics Application, was developed to assist residents, consumers, public market stall owners in tracking the prices of the commodities in District 2, which consists of Cabuyao City, Los Baños and Bay. The system provides essential features like market data and price tracking, price forecasting, product browsing and search, custom location selection, seller functions for sellers in the market, and user experience.

The development of the system aimed to address the pressing issues of inflation and the unfair market prices of some stall owners. The researchers utilized web technologies to ensure accessibility and responsiveness across various devices. This strategic choice was fundamental to promoting market transparency, as it is guaranteed that both consumers and small market vendors could access real-time price data instantly. The system's architecture was further enhanced by the inclusion of specialized modules for Administrators and Users a structural decision that amplified its ability to enforce fairness.

An ISO-based system evaluation was conducted covering functional suitability, performance efficiency, compatibility, usability, reliability, and security. The results showed that users rated the system as Very Good to

Excellent in all categories, indicating that “Price Pulse” effectively met its objectives and user expectations.

Conclusion

The development of the web-based and mobile-based system Price Pulse has proven to be a significant advancement in the field of the Department of Agriculture. By developing the Price Pulse, the system offers an accessible and efficient tool for monitoring the price of every product in the market of District 2. In conclusion, the Price Pulse system undergoes development though the following objectives.

1. Develop a multi-platform application using web-based and mobile-based application that provides the real-time price of agricultural and poultry products.

The researchers conducted a comprehensive data gathering process focused on the system that provides real-time price of agricultural and poultry products. The researchers created the multi-platform application, encompassing both web and mobile systems, successfully achieves the goal of delivering real-time price tracking for agricultural and poultry goods in Laguna’s District 2. The researchers incorporate the elements like data entry, administrative verification, data visualization analytics, and predictive modeling, the system guarantees precision, clarity, and availability for vendors and consumers. It closes the gap in market knowledge by providing trustworthy and current pricing information that promotes fair trading and informed choices. The use of predictive analytics enables users to foresee market changes and prepare accordingly. The initiative demonstrates that technological progress

can improve market transparency, strengthen local economies, and promote sustainable agricultural development.

2. Integrate data analytics features that visualize price trends, predict market fluctuations, and offer insights for better decision-making in agricultural trade.

The system is successfully incorporated data analytics tool that help users understand and interpret market data more effectively. Through the integration of tools that display pricing patterns, the system facilitates the viewing and analysis of changes in agricultural prices over specified time periods by users, including sellers and buyers. Visual aids for studying and comprehending complex price data include charts, graphs, dashboards, and cards. Users can examine historical and present prices, spot trends, forecast potential market movements, and track seasonal variations with these graphic tools. This enhances clarity and allows customers to quickly make informed choices by leveraging both past and present pricing information. The predictive models improve the system's ability to anticipate market changes by examining past data and external elements that affect agricultural prices. These models, which are based on data analysis techniques, can provide accurate projections that enable users to more precisely predict future price swings. This lowers the danger of abrupt changes in the market by enabling suppliers and retailers to better plan their production, sales, and inventory management.

In addition, the system provides comprehensive analytics and data-driven interpretations, transforming raw data into information that supports strategic decision-making. The ideal times to buy, sell, or preserve agricultural

products can be determined by stakeholders using these data to optimize earnings and minimize losses. The agriculture market is more competitive as a result of this analytical and predictive ability.

3. Evaluate the system's performance in terms of accuracy, usability, and reliability through user testing, feedback collection, and comparative analysis with existing price monitoring methods.

Through comprehensive user testing, feedback collection, and comparison with existing price monitoring methods, the system's overall efficacy was assessed based on accuracy, usability, and reliability. The algorithm produced and predicted prices with a high degree of precision, according to the results, however there were minor deviations from actual market numbers. Additionally, user comments indicated that the system's design was easy to use and straightforward, which improved the user experience overall. Furthermore, the developed system demonstrated improved efficacy and dependability in monitoring and updating pricing data in comparison to existing methods. These results confirm that the system accomplishes its objectives and provides a helpful tool for monitoring and assessing prices in real time.

Recommendation

Based on the findings of this project, the following are highly recommended.

1. Future researchers are encouraged to adopt more advanced predictive models or integrate AI techniques such as machine learning or deep

learning. These approaches can analyze larger datasets, uncover hidden patterns, and produce more precise forecasts.

2. Predictive models can be further improved by incorporating external factors like natural disasters, animal diseases, pest infestations, fuel costs, government policies, and taxation. Accounting for both environmental and economic variables create a more comprehensive and adaptive framework that supports strategic planning and decision-making.
3. It is recommended to develop a mobile app for administrators to monitor and manage the system remotely. This innovation allows admins to oversee operations even when away from the office, enhancing flexibility and efficiency.
4. Adding feedback or reporting feature would enable users to report suspicious or incorrect price entries. This promotes data accuracy, transparency, and user participation while helping detect errors or price manipulation more effectively.
5. To support long-term adoption, the system should be upgraded to comply with Department of Agriculture (DA) standards. This includes aligning data structures, using standardized reporting formats, and ensuring interoperability for official public deployment.

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APPENDICES

Appendix A
Letter of Approval

APPROVAL SHEET

The Capstone Project entitled "**REAL-TIME PRICE MONITORING AND MARKET PRICE PREDICTION FOR AGRICULTURAL COMMODITIES IN DISTRICT 2: A MULTI PLATFORM DATA ANALYTICS APPLICATION**" prepared and submitted by **Renz Leonard A. Guevarra, John Vincent V. Sorino, and Jilian Mar M. Sierra** in partial fulfillment requirements for the degree Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development is hereby recommended for approval and acceptance.

JEFFERSON L. LERIOS
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of____.

MIKE PHILIP T. RAMOS
Member

CAROLINA R. JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

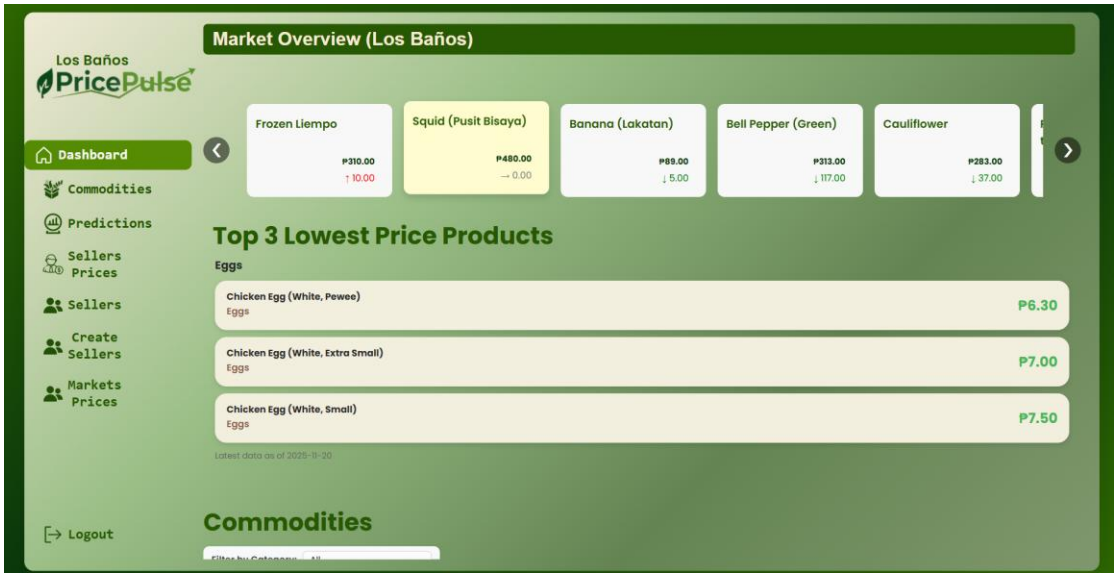
OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

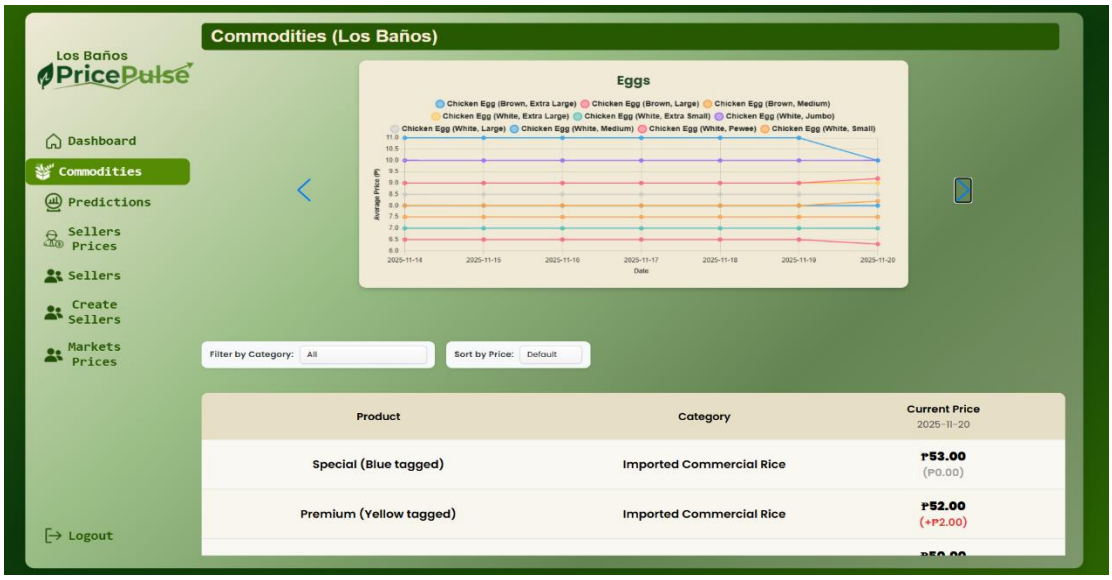
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Date Signed

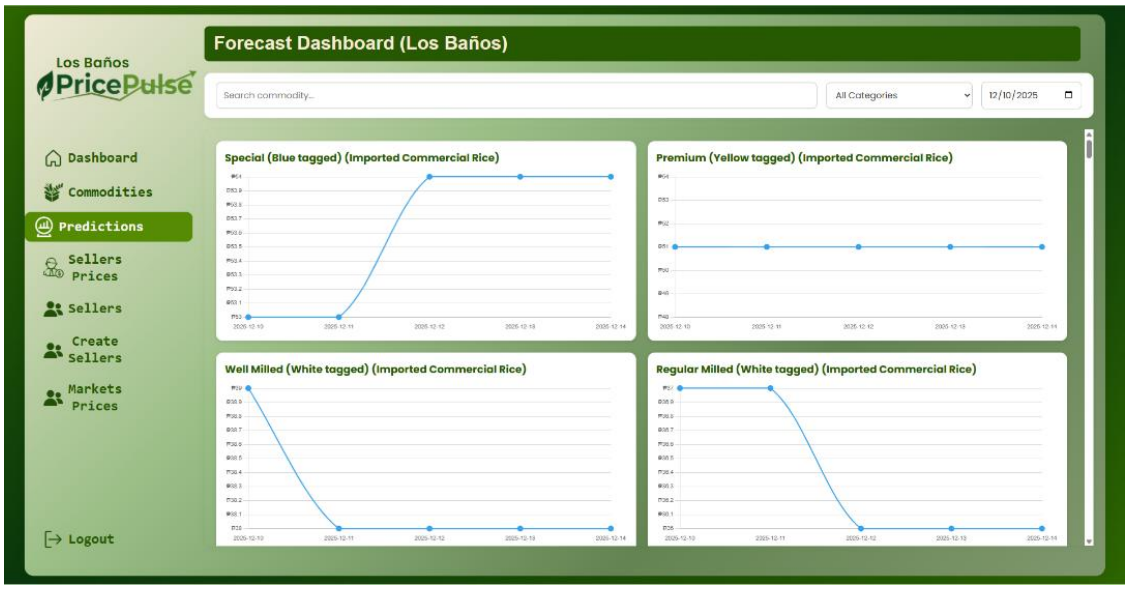
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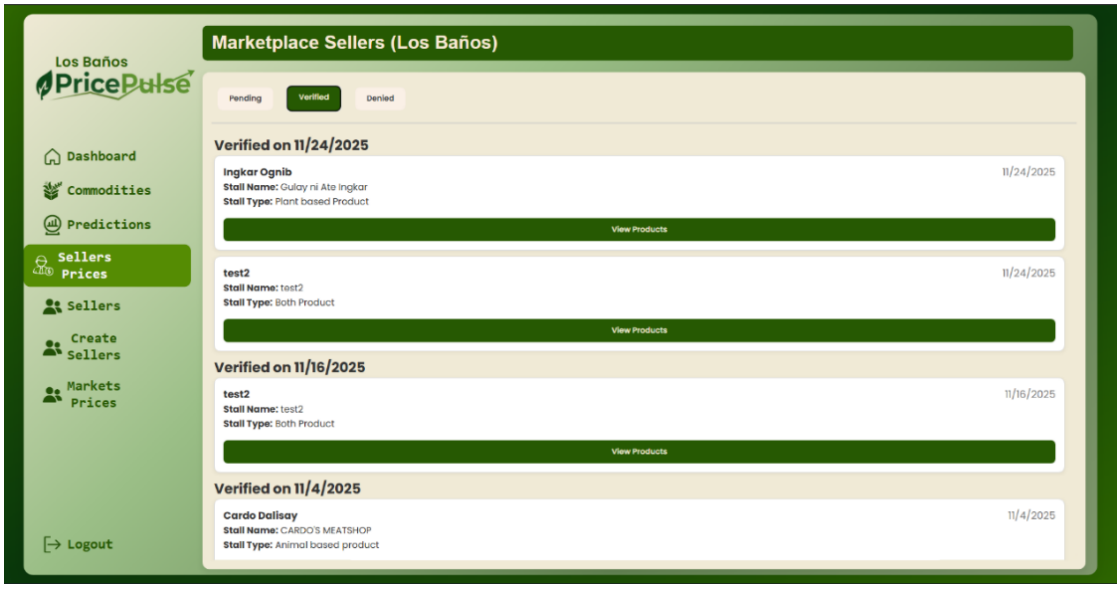
Appendix B

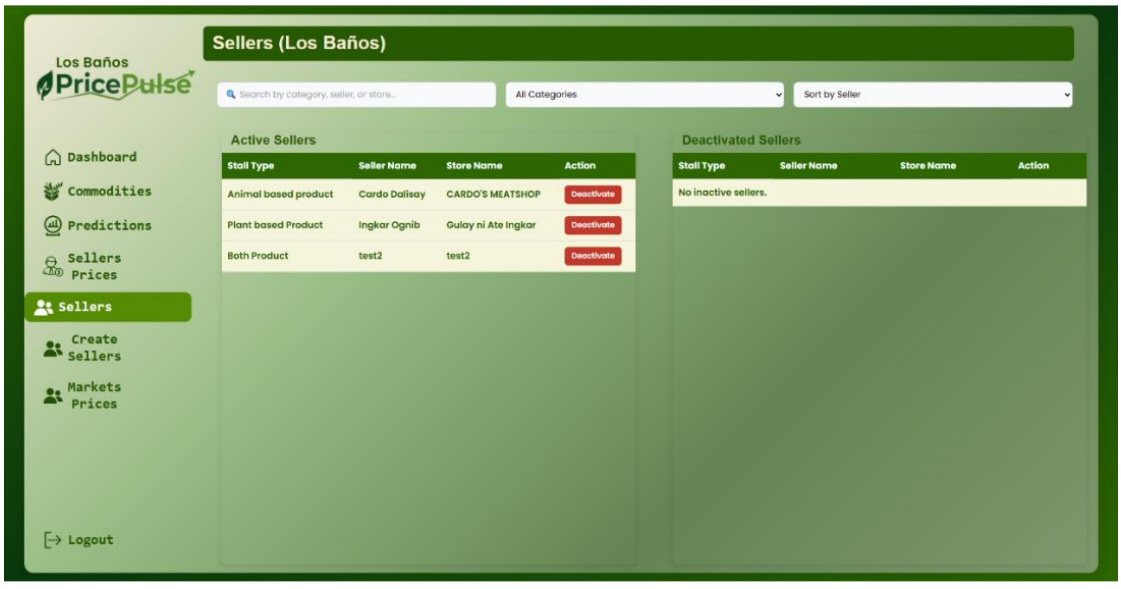
System Interface Screenshots

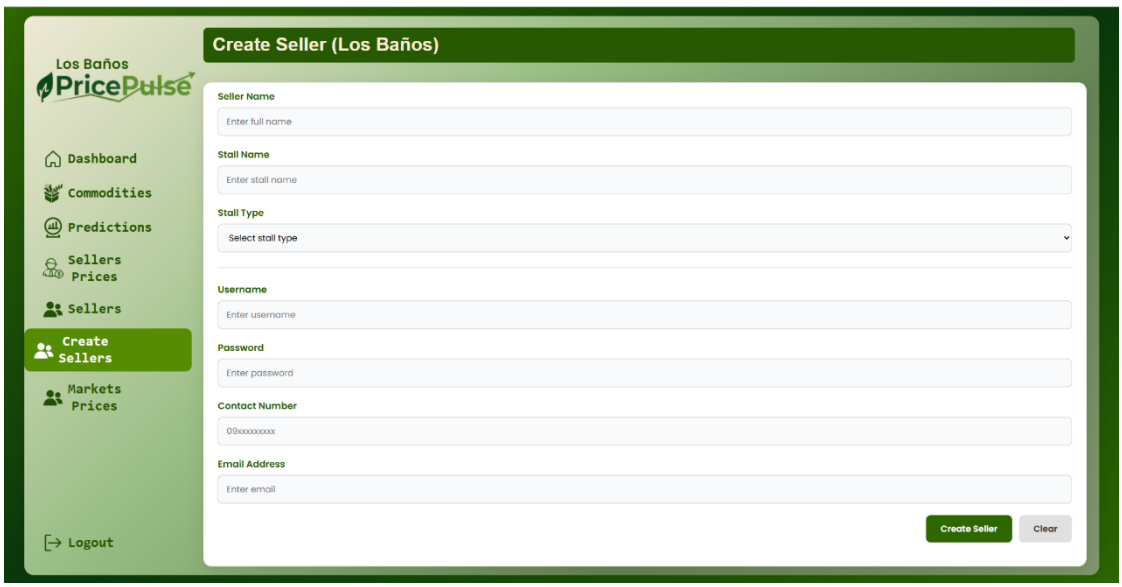

Los Baños PricePulse
Market Overview (Los Baños)
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout
Frozen Liempo
₱310.00
↑ 10.00
Squid (Pusit Bisaya)
₱480.00
→ 0.00
Banana (Lakatan)
₱89.00
↓ 5.00
Bell Pepper (Green)
₱313.00
↓ 117.00
Cauliflower
₱283.00
↓ 37.00
Top 3 Lowest Price Products
Eggs
Chicken Egg (White, Pewee)
Eggs
₱6.30
Chicken Egg (White, Extra Small)
Eggs
₱7.00
Chicken Egg (White, Small)
Eggs
₱7.50
Latest data as of 2025-11-20
Commodities

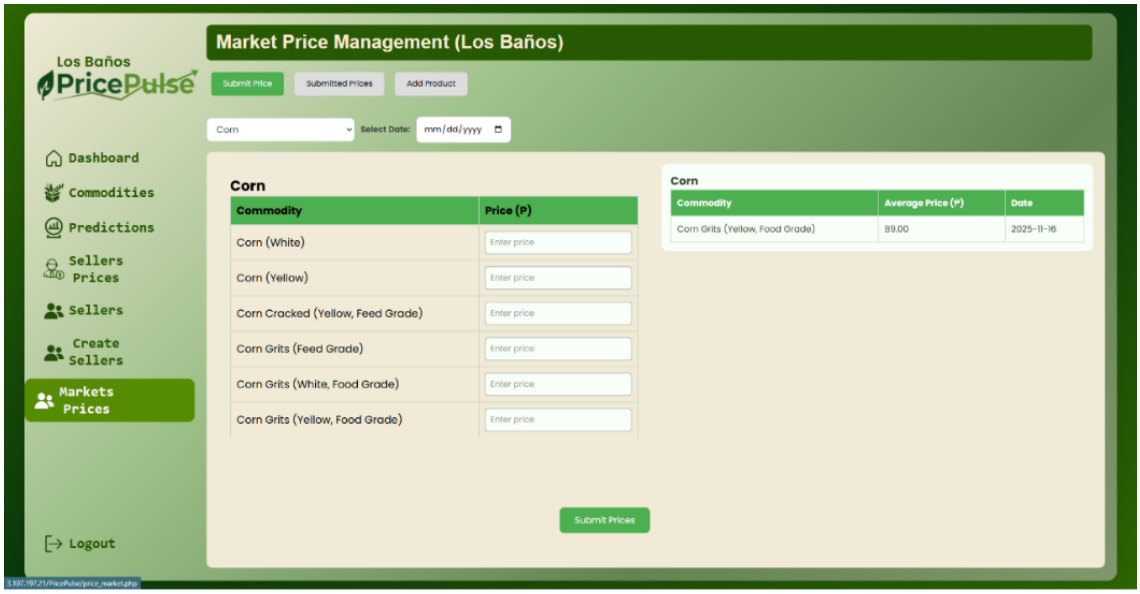

Los Baños PricePulse
Commodities (Los Baños)
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout
Eggs
Chicken Egg (Brown, Extra Large)
Chicken Egg (Brown, Large)
Chicken Egg (Brown, Medium)
Chicken Egg (White, Extra Large)
Chicken Egg (White, Extra Small)
Chicken Egg (White, Jumbo)
Chicken Egg (White, Large)
Chicken Egg (White, Medium)
Chicken Egg (White, Pewee)
Chicken Egg (White, Small)
Average Price (₱)
Date
Filter by Category: All
Sort by Price: Default
Product
Category
Current Price
2025-11-20
Special (Blue tagged)
Imported Commercial Rice
₱53.00
(₱0.00)
Premium (Yellow tagged)
Imported Commercial Rice
₱52.00
(+₱2.00)

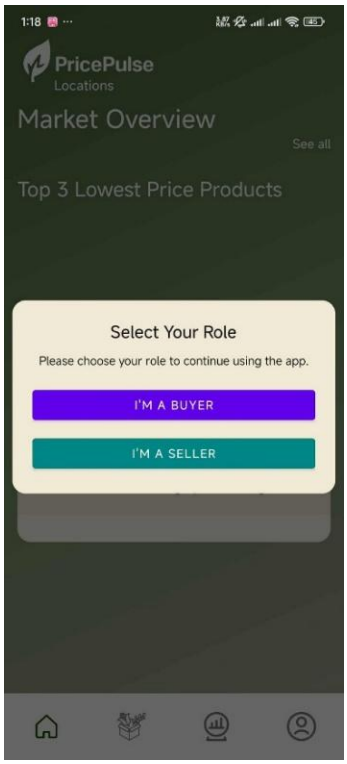

Los Baños PricePulse
Forecast Dashboard (Los Baños)
Search commodity...
All Categories
12/10/2025
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout
Special (Blue tagged) (Imported Commercial Rice)
Premium (Yellow tagged) (Imported Commercial Rice)
Well Milled (White tagged) (Imported Commercial Rice)
Regular Milled (White tagged) (Imported Commercial Rice)

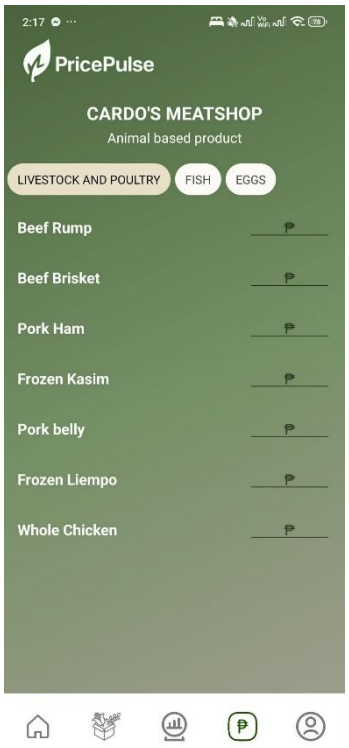

Los Baños PricePulse
Marketplace Sellers (Los Baños)
Pending
Verified
Denied
Verified on 11/24/2025
Ingkar Ognib
Stall Name: Gulay ni Ate Ingkar
Stall Type: Plant based Product
11/24/2025
View Products
test2
Stall Name: test2
Stall Type: Both Product
11/24/2025
View Products
Verified on 11/16/2025
test2
Stall Name: test2
Stall Type: Both Product
11/16/2025
View Products
Verified on 11/4/2025
Cardo Dalisay
Stall Name: CARDO'S MEATSHOP
Stall Type: Animal based product
11/4/2025
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout

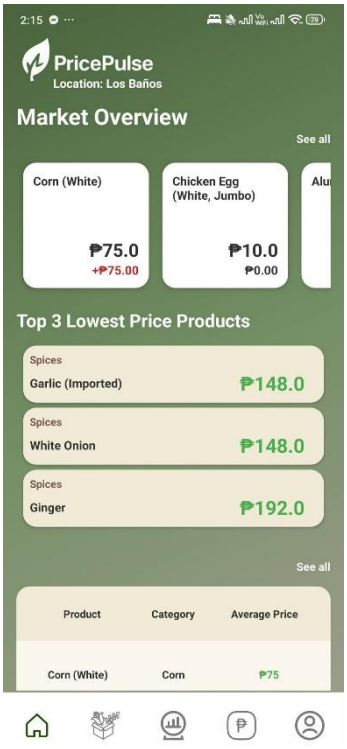

Los Baños PricePulse
Sellers (Los Baños)
Search by category, seller, or store...
All Categories
Sort by Seller
Active Sellers
Stall Type
Seller Name
Store Name
Action
Animal based product
Cardo Dalisay
CARDO'S MEATSHOP
Deactivate
Plant based Product
Ingkar Ognib
Gulay ni Ate Ingkar
Deactivate
Both Product
test2
test2
Deactivate
Deactivated Sellers
Stall Type
Seller Name
Store Name
Action
No inactive sellers.
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout

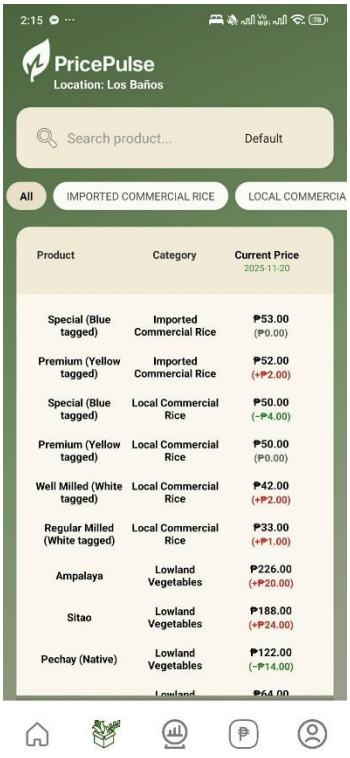

Los Baños PricePulse
Create Seller (Los Baños)
Seller Name
Enter full name
Stall Name
Enter stall name
Stall Type
Select stall type
Username
Enter username
Password
Enter password
Contact Number
09xxxxxxxxx
Email Address
Enter email
Create Seller
Clear
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout


Los Baños
PricePulse
Market Price Management (Los Baños)
Submit Price
Submitted Prices
Add Product
Corn
Select Date:
mm/dd/yyyy
Dashboard
Commodities
Predictions
Sellers Prices
Sellers
Create Sellers
Markets Prices
Logout
Corn
Commodity
Price (₱)
Corn (White)
Corn (Yellow)
Corn Cracked (Yellow, Feed Grade)
Corn Grits (Feed Grade)
Corn Grits (White, Food Grade)
Corn Grits (Yellow, Food Grade)
Enter price
Submit Prices
Corn
Commodity
Average Price (₱)
Date
Corn Grits (Yellow, Food Grade)
89.00
2025-11-16

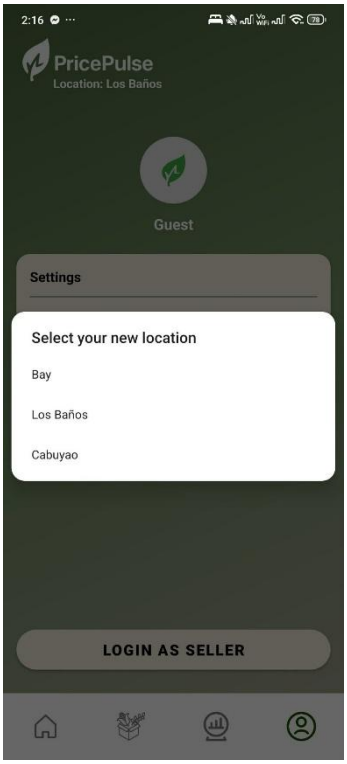

PricePulse
Locations
Market Overview
See all
Top 3 Lowest Price Products
Select Your Role
Please choose your role to continue using the app.
I'M A BUYER
I'M A SELLER


PricePulse
CARDO'S MEATSHOP
Animal based product
LIVESTOCK AND POULTRY
FISH
EGGS
Beef Rump
Beef Brisket
Pork Ham
Frozen Kasim
Pork belly
Frozen Liempo
Whole Chicken


PricePulse
Location: Los Baños
Market Overview
See all
Corn (White)
₱75.0
+₱75.00
Chicken Egg (White, Jumbo)
₱10.0
₱0.00
Top 3 Lowest Price Products
Spices
Garlic (Imported)
₱148.0
Spices
White Onion
₱148.0
Spices
Ginger
₱192.0
See all
Product
Category
Average Price
Corn (White)
Corn
₱75


PricePulse
Location: Los Baños
Search product...
Default
All
IMPORTED COMMERCIAL RICE
LOCAL COMMERCIA
Product
Category
Current Price
2025-11-20
Special (Blue tagged)
Imported Commercial Rice
₱53.00
(₱0.00)
Premium (Yellow tagged)
Imported Commercial Rice
₱52.00
(+₱2.00)
Special (Blue tagged)
Local Commercial Rice
₱50.00
(−₱4.00)
Premium (Yellow tagged)
Local Commercial Rice
₱50.00
(₱0.00)
Well Milled (White tagged)
Local Commercial Rice
₱42.00
(+₱2.00)
Regular Milled (White tagged)
Local Commercial Rice
₱33.00
(+₱1.00)
Ampalaya
Lowland Vegetables
₱226.00
(+₱20.00)
Sitao
Lowland Vegetables
₱188.00
(+₱24.00)
Pechay (Native)
Lowland Vegetables
₱122.00
(−₱14.00)


PricePulse
Location: Los Baños
Search product...
All
IMPORTED COMMERCIAL RICE
LOCAL COMME
SELECT DATE
2025-11-23
Product
Category
Forecast Price
Corn (White)
Corn
₱70
Corn (Yellow)
Corn
₱70
Corn Grits (White, Food Grade)
Corn
₱60
Chicken Egg (Brown, Extra Large)
Eggs
₱11
Chicken Egg (Brown, Large)
Eggs
₱9
Chicken Egg (Brown, Medium)
Eggs
₱8
Chicken Egg


PricePulse
Location: Los Baños
Guest
Settings
Select your new location
Bay
Los Baños
Cabuyao
LOGIN AS SELLER

Appendix C

Relevant Source Code

```
import pandas as pd
import requests
import numpy as np
from sqlalchemy import create_engine
from neuralprophet import NeuralProphet

# ------------------------------------------------------------
# Weather API Fetcher
# ------------------------------------------------------------
def fetch_weather_data(lat, lon, start_date, end_date, is_historical=True):
    if is_historical:
        url = "https://archive-api.open-meteo.com/v1/archive"
    else:
        url = "https://api.open-meteo.com/v1/forecast"

    params = {
        "latitude": lat,
        "longitude": lon,
        "start_date": start_date,
        "end_date": end_date,
        "daily": [
            "precipitation_sum",
            "wind_speed_10m_max",
            "temperature_2m_max",
            "temperature_2m_min",
        ],
        "timezone": "Asia/Manila"
    }

    try:
```

```
        response = requests.get(url, params=params)
        response.raise_for_status()
        d = response.json()["daily"]

        return pd.DataFrame({
            "ds": pd.to_datetime(d["time"]),
            "rainfall": d["precipitation_sum"],
            "wind_speed_max": d["wind_speed_10m_max"],
            "temp_max": d["temperature_2m_max"],
            "temp_min": d["temperature_2m_min"],
        })

    except Exception as e:
        print("Weather API Error:", e)
        return pd.DataFrame()

# ------------------------------------------------------------
# Database Connection
# ------------------------------------------------------------
User = 'admin'
Password = 'Tanso_25'
Host = 'pricepulse.clgiq4q2cm4p.ap-southeast-2.rds.amazonaws.com'
Database = 'pricepulse_db'
engine =
create_engine(f"mysql+pymysql://{User}:{Password}@{Host}/{Database}")

query = """
    SELECT
        cp.date AS ds,
        cp.average_price AS y,
        cp.commodity_id,
```

```
        c.name AS commodity_name,
        cp.location_id,
        l.location_name
    FROM commodity_prices_tbl cp
    JOIN commodity_tbl c ON cp.commodity_id = c.commodity_id
    JOIN location_tbl l ON cp.location_id = l.location_id
    ORDER BY cp.date ASC;
    """

df = pd.read_sql(query, engine)
df["ds"] = pd.to_datetime(df["ds"])

print("Loaded rows:", len(df))
# ------------------------------------------------------------
# Weather Data
# ------------------------------------------------------------
LAT = 14.1721
LON = 121.2431
PRED_DAYS = 14

hist_start = df["ds"].min().strftime("%Y-%m-%d")
hist_end = df["ds"].max().strftime("%Y-%m-%d")

future_start = (df["ds"].max() + pd.Timedelta(days=1)).strftime("%Y-%m-%d")
future_end = (df["ds"].max() +
pd.Timedelta(days=PRED_DAYS)).strftime("%Y-%m-%d")

print("\nFetching historical weather...")
hist_weather = fetch_weather_data(LAT, LON, hist_start, hist_end, True)

print("Fetching future weather...")
```

```
future_weather = fetch_weather_data(LAT, LON, future_start, future_end, False)


weather_all = pd.concat([hist_weather, future_weather]).drop_duplicates("ds")


# ------------------------------------------------------------

# Merge price + weather

# ------------------------------------------------------------

df_train = pd.merge(df, weather_all, on="ds", how="inner")

df_train["unique_id"] = df_train["location_id"].astype(str) + "_" + df_train["commodity_id"].astype(str)

df_train.sort_values(["unique_id", "ds"], inplace=True)

df_train.loc[df_train["y"] <= 0, "y"] = np.nan

df_train["y"] = df_train.groupby("unique_id")["y"].transform(lambda x: x.ffill())

df_train.dropna(subset=["y"], inplace=True)


# ------------------------------------------------------------

# Prediction for each commodity

# ------------------------------------------------------------

db_forecast_records = []

group_counter = 0


for uid, group in df_train.groupby("unique_id"):

    group_counter += 1

    commodity_name = group["commodity_name"].iloc[0]

    location_name = group["location_name"].iloc[0]


    print(f"\n ◘◘◘ Processing Group {group_counter}: {commodity_name} @ {location_name} ({uid}) with {len(group)} records ◘◘◘")


    if len(group) < 14:
```

```
        print("   <!> Skipping (not enough data).")
        continue

    group = group.sort_values("ds").reset_index(drop=True)

    try:
        # Select only allowed NeuralProphet columns
        group_data = group[["ds", "y", "temp_max", "temp_min", "rainfall",
"wind_speed_max"]].copy()

        if group_data["y"].nunique() < 2:
            print("   <!> Skipping (constant values).")
            continue

        # ------------------------------
        # Model Setup
        # ------------------------------
        model = NeuralProphet(
            n_changepoints=20,
            yearly_seasonality=True,
            weekly_seasonality=True,
            daily_seasonality=False
        )

        model.add_future_regressor("temp_max")
        model.add_future_regressor("temp_min")
        model.add_future_regressor("rainfall")
        model.add_future_regressor("wind_speed_max")

        model.fit(group_data, freq="D")
```

```
future_df = model.make_future_dataframe(
    group_data,
    periods=PRED_DAYS,
    regressors_df=future_weather
)

forecast = model.predict(future_df)

# ----------------------------------
# Attach metadata back to forecast
# ----------------------------------
forecast["commodity_id"] = group["commodity_id"].iloc[0]
forecast["location_id"] = group["location_id"].iloc[0]
forecast["unique_id"] = uid
forecast["commodity_name"] = commodity_name
forecast["location_name"] = location_name

forecast["yhat1"] = forecast["yhat1"].round(0).astype(int)

db_rows = pd.DataFrame({
"commodity_id": int(group["commodity_id"].iloc[0]),
"location_id": int(group["location_id"].iloc[0]),
"forecast_date": pd.to_datetime(forecast["ds"]).dt.date,
"predicted_price": forecast["yhat1"].round(0).astype(float),
"created_at": pd.Timestamp.now().date()
})

db_rows = db_rows.astype({
"commodity_id": "int64",
"location_id": "int64",
```

```
        "predicted_price": "float64"
        })

        db_rows["forecast_date"] =
pd.to_datetime(db_rows["forecast_date"]).dt.date
        db_rows["created_at"] = pd.to_datetime(db_rows["created_at"]).dt.date

        db_forecast_records.append(db_rows)

        try:
            db_rows.to_sql(
                name="commodity_forecast_tbl",
                con=engine,
                if_exists="append",
                index=False
            )
            print(f"✓ Inserted {len(db_rows)} rows for UID {uid}")

        except Exception as e:
            print(f"<!> SQL Insert Error for UID {uid}: {e}")
            engine.dispose()

        print(f"Saved forecast for {uid}")

    except Exception as e:
        print(f" <!> Error processing {uid}: {e} <!> ")
```

Appendix D

Evaluation Tool

Real-Time Price Monitoring and Market Prediction for Agricultural Commodities in District 2: A Multi-Platform Data Analytics Application

Renz Leonard Guevarra

John Vincent V. Sorino

Jilian Mar M. Sierra

Name of Respondents(optional):_____

ISO25010 Survey Questionnaire

Sub-Criteria	Description / Indicators	1 – Very Low	2 – Low	3 – Moderate	4 – High
I.	**Functional Sustainability**				
Essential Functions	The application provides all necessary functions for its intended purpose.	Missing essential functions; fails to meet objectives.	Some core functions are incomplete or unstable.	Most essential functions are operational.	All required functions are complete and reliable.
Feature Accuracy	Each feature performs accurately according to specified requirements.	Frequent feature errors and failures.	Features function inconsistently.	Features perform as intended with minor issues.	All features perform accurately as specified.
Consistency of Results	The system delivers consistent outputs across various functions.	Results are inconsistent and unreliable.	Partial consistency in results.	Results are mostly consistent.	Results are fully consistent and reliable.
II.	**Performance Efficiency**				
Response Time	The system responds quickly within acceptable time limits.	System is slow and unresponsive.	Response time often delayed.	Generally performs within acceptable limits.	Responds quickly and efficiently.
Resource Utilization	The application efficiently uses CPU and memory resources.	Consumes excessive resources.	Moderate inefficiency in resource use.	Resources adequately managed.	Optimized resource use across operations.

Stability Under Load	The system maintains performance with heavy data or users.	Crashes or slows under load.	Reduced performance during heavy usage.	Performs acceptably under moderate load.	Stable and efficient even under high load.
Benchmark Compliance	The performance meets expected test benchmarks.	Fails to meet most benchmarks.	Meets some benchmarks only.	Meets most benchmarks.	Exceeds all benchmarks expectations.
III. Compatibility					
Platform Compatibility	Operates effectively across Android versions and devices.	Works on limited devices only.	Inconsistent across versions.	Runs on most Android versions.	Fully compatible with all devices/versions.
System Integration	Integrates with external software or data sources.	Integration often fails.	Limited integration capability.	Integrates with some systems.	Seamless integration with all systems.
Third-party Interaction	Maintains function with third-party tools or APIs.	Fails with most third-party tools.	Limited compatibility with APIs.	Works with common third-party tools.	Fully compatible and stable with APIs.
Interoperability	Supports operation across various environments.	Operates only in restricted setups.	Limited interoperability.	Works across several environments.	Fully interoperable across platforms.
IV. Usability					
Interface Design	Supports intuitive interaction and easy learning.	Confusing and difficult interface.	Requires effort to learn.	Interface is understandable with some guidance.	Very intuitive and user-friendly.
User Feedback	Provides meaningful feedback and guidance.	No or unclear feedback.	Feedback is limited or delayed.	Provides adequate guidance.	Clear, helpful, and responsive feedback.
Navigation & Controls	Icons, controls, and	Inconsistent and	Some inconsist	Generally consisten	Fully consistent

	navigation are consistent.	hard to locate.	encies in layout.	t and usable.	and easy to navigate.
V.	**Readability**				
Code Structure	Code is well-organized and documented.	Disorganized and undocumented.	Partial documentation.	Generally structured and understandable.	Well-structured and fully documented.
Naming & Comments	Naming conventions and comments are clear.	Unclear or missing comments.	Inconsistent naming.	Mostly clear with minor gaps.	Consistent and descriptive naming/comments.
Logic Flow	Logic is easy to trace and maintain.	Difficult to follow.	Some unclear logic paths.	Traceable with moderate effort.	Clear, logical, and easy to maintain.
Documentation Accuracy	Documentation matches implementation.	Outdated or missing.	Some mismatches.	Generally accurate with few errors.	Fully up-to-date and accurate.
VI.	**Security**				
Data Protection	Protects user data from unauthorized access.	Frequent data breaches.	Limited data protection.	Adequate protection with minor risks.	Strong data protection and encryption.
Authentication & Encryption	Uses secure authentication methods.	No encryption.	Weak protocols.	Basic security measures implemented.	Robust encryption and authentication systems.
Vulnerability Testing	Tests and mitigates vulnerabilities.	Rarely tested or patched.	Occasional checks.	Regular testing and partial fixes.	Continuous testing and full mitigation.
Compliance Standards	Follows privacy/security standards.	Non-compliant.	Partially compliant.	Meets most security regulations.	Fully compliant with all security standards.
VII.	**Maintainability**				
Code Modifiability	Supports easy modification and debugging.	Difficult to modify.	Minor improvements possible.	Relatively easy to modify.	Easily modifiable and debuggable.

Update Management	Updates can be applied with minimal disruption.	Updates cause major issues.	Require downtime.	Updates manageable with some effort.	Smooth and non-disruptive updates.
Future Enhancement	Design supports future upgrades.	Not scalable.	Limited enhancement capability.	Allows moderate feature expansion.	Highly adaptable for future improvements.
Coding Standards	Follows standard coding practices.	No standards used.	Inconsistent practices.	Mostly follows coding standards.	Fully adheres to best practices.
VIII. Portability					
Platform Deployment	Easily deployed across platforms/environments.	Deployment fails.	Requires major adjustments.	Deployable with minor setup.	Easily deployable on multiple platforms.
Adaptability	Easily adapted to new configurations.	Requires complete reconfiguration.	Adapts with modifications.	Moderate effort to adapt.	Quickly adaptable to new setups.
Dependency Management	Dependencies clearly defined and manageable.	Unclear dependencies.	Some issues.	Dependencies mostly managed.	All dependencies clearly defined and stable.
Installation & Migration	Supports smooth installation and migration.	Installation fails.	Complex steps required.	Generally smooth process.	Simple, reliable installation and migration.

Real-Time Price Monitoring and Market Prediction for Agricultural Commodities in District 2: A Multi-Platform Data Analytics Application

Renz Leonard Guevarra

John Vincent V. Sorino

Jilian Mar M. Sierra

Name of Respondents(optional):_____

Perceived Usefulness	**5**	**4**	**3**	**2**	**1**
1. The application helps me efficiently monitor product price changes.					
2. Using the application improves my ability to make informed purchasing decisions.					
3. The application provides accurate and relevant information that is beneficial to me.					
4. The application enhances my productivity when checking product prices.					
Perceived Ease of Use					
1. Learning how to use the application was easy for me.					
2. It is simple to find and access the features I need within the application.					
3. The layout and design of the application make it easy to understand.					
4. I can use the application effectively without needing technical assistance.					
User Satisfaction					
1. I am satisfied with the overall performance of the application.					
2. The application meets my expectations in terms of functionality and reliability.					
3. I am pleased with how quickly and smoothly the application responds.					
4. I would recommend this application to others because of my positive experience.					
Willingness to Adopt					
1. I am likely to continue using the application feature.					
2. I would prefer this application over other similar tools.					
3. I am willing to rely on this application for checking product prices regularly.					
4. I am interested in using future updates or improved versions of this application.					

Appendix E

Curriculum Vitae

John Vincent V. Sorino

- 0927-218-7371
- sorinojv@gmail.com
- #841 Palupit Compd. Brgy. Real Calamba Laguna



EXPERIENCE

Upwork Freelance
September 2024 - March 2025
- Graphic Design
- Game Testing
- AI-Training and Validation

Freelance graphic designer with a strong foundation in gaming and design. Proficient in creating stunning visuals, logos, advertisements, and photo editing using Photoshop and Figma. Skilled in both digital art and traditional drawing, with knowledge of programming language like Python, C#, and Java. Dedicated to providing creative and practical solutions for clients.

EDUCATION

College
- Bachelor of Science in Information Technology
- Laguna State Polytechnic University
- Brgy. Malinta, Los Baños
- 2022 - Present

PERSONAL INFORMATION

Date of Birth: April 05, 2002
Place of Birth: Real Calamba City
Gender: Male
Civil Status: Single

SKILLS & EXPERTISE

Ability to work clean
Strong communication skills
Willing to take risk and take new challenges
Willingness to assist teammates when needed.
Programming
Proficiency in Microsoft Word & Excel
Canva
Figma
Photoshop



Brgy. Sto. Domingo,
Bay, Laguna 0433
09107836058
Guevarra.leonard25@gmail.com

RENZ LEONARD GUEVARRA

summary

Hardworking, highly motivated professional eager to lend combined knowledge and skills to enhance business performance. Operates well in both individual and team capacities, leveraging seasoned work ethic to quickly adapt to different processes and drive company objectives. Resourceful and results-driven with a passion for growth and efficiency to meet company needs and increase service value..

Skills & Abilities

Proficient in programming languages such as Java, and PHP

Familiar with web development technologies such as HTML, CSS, JavaScript

Knowledgeable in database management using MySQL and Oracle

Experience in using Git and version control tools

Strong analytical and problem-solving skills

Excellent verbal and written communication skills

Ability to work effectively in a team environment

Education

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY, LAGUNA STATE POLYTECHNIC UNIVERSITY – LOS BAÑOS, JUNE 2026

References

REFERENCES ARE AVAILABLE UPON REQUEST. PLEASE FEEL FREE TO CONTACT ME FOR ANY FURTHER INFORMATION.

JILIAN MAR M. SIERRA

jilianmarsierra@gmail.com | 09353022157 | 0085 Lalakay, Los Baños, Laguna

SUMMARY

IT student with experience in Python-based development, data systems, and IT operations. Technical Lead for a real-time price monitoring system with forecasting. Strong background in troubleshooting, hardware support, and system management.

TECHNICAL PROJECTS

Price Pulse (Real-time Price Monitoring System for Agricultural Commodities) | Technical Lead & Backend / Data Developer **Aug 2025 - Dec 2025**

- Developed Python-based price prediction models using NeuralProphet for time-series forecasting, incorporating weather data as external factors.
- Design and managed a relational database storing 65,000+ historical and real-time commodity price records.
- Worked with AWS infrastructure to configure server environments, manage databases, and support backend application execution.
- Automated data processing and model execution using Linux crontab to schedule Python scripts on the AWS server.
- Defined system features and requirements for both user and admin roles, guiding backend development for web and mobile systems.

WORK EXPERIENCE

Computer Cafe Assistant | Julia's Internet Shop - Los Baños, Laguna **Oct 2018 - Dec 2019**

- Managed daily operations of a computer café, ensuring stable system performance and customer support.
- Performed basic hardware and software troubleshooting for desktops, peripherals, and network connectivity issues.
- Assisted customers with printing, document typing, and file handling tasks.
- Maintained computer units through routine checks, software setup, and minor repairs.
- Handled customer transactions and provided technical assistance in a fast-paced environment.

Airconditioning Technician (Part-Time) | JMRC Airconditioning Service - Laguna **Apr 2020 - Present**

- Performed preventive maintenance, cleaning, repair, and installation of air conditioning units for commercial and residential clients.
- Serviced large corporate and institutional clients, including manufacturing plants such as Torica and Bayer, as well as banks and restaurant establishments across multiple locations.
- Conducted dismantling, relocation, and reinstallation of air conditioning systems following safety and quality standards.
- Coordinated with team members to complete service tasks efficiently and professionally.

Data Entry Clerk | Confidential Client **Sept 2024 - Mar 2025**

- Extracted and processed insurance-related data from internal company systems.
- Entered and validated data using Microsoft Excel, ensuring accuracy and consistency.
- Organized datasets and submitted structured reports back into the company system.
- Maintained data confidentiality and followed company data handling procedures.

EDUCATION

Bachelor of Science in Information Technology Major in Web and Mobile Development **2022 - 2026**
Laguna State Polytechnic University - Los Baños, Laguna

CERTIFICATIONS & AWARDS

- **3rd Place -** Programming: Code Wars - LSPU LB (2024)
- **AI Literacy** - IBM SkillsBuild (2025)
- **Cybersecurity Fundamentals** - IBM SkillsBuild (2025)
- **Data Fundamentals** - IBM SkillsBuild (2025)
- **Digital Literacy** - IBM SkillsBuild (2025)
- **Fundamentals of Sustainability and Technology** - IBM SkillsBuild (2025)
- **Getting Started with Cybersecurity** - IBM SkillsBuild (2025)
- **Getting Started with Data -** IBM SkillsBuild (2025)
- **Getting Started with Generative AI** - IBM SkillsBuild (2025)
- **Information Technology Fundamentals** - IBM SkillsBuild (2025)
- **Machine Learning with Python** - IBM SkillsBuild (2025)
- **Python for Data Science -** IBM SkillsBuild (2025)
- **Web Development Fundamentals** - IBM SkillsBuild (2025)
- **Working in a Digital World Professional Skills** - IBM SkillsBuild (2025)

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, SQL, C#, PHP
- **Tools and Technologies:** AWS, Anaconda, Crontab, Linux, NeuralProphet
- **Design Tools:** Figma, Adobe Photoshop, Canva
- **Soft Skills:** Analytical Thinking, Technical Communication, Problem Solving

Appendix F

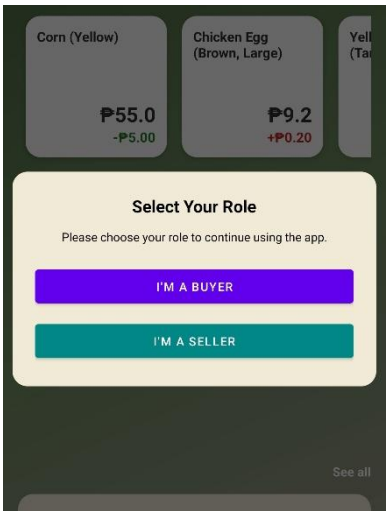
User's Guide



USER MANUAL

Selecting Your Role

When you first open the app, you will be prompted to select your role. This choice determines the initial screens and features you will access

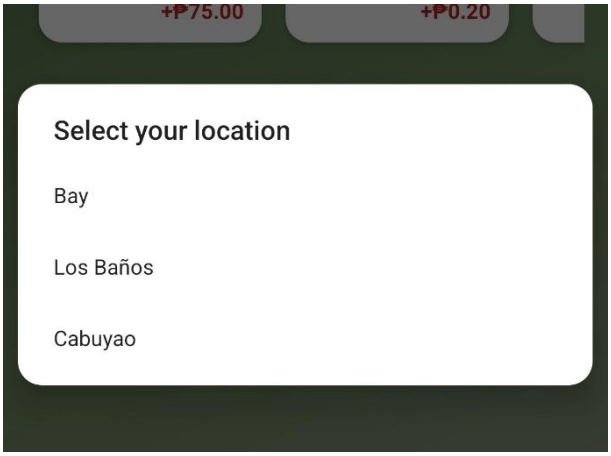


Navigation Flow Details

A. Buyer Flow (I'M A BUYER)

If you select **I'M A BUYER**, the app assumes you want to immediately start browsing available products:

1. Tap the **I'M A BUYER** (Purple) button.
2. You will be directed to the **Select Location** screen where you will be asked to choose your location.



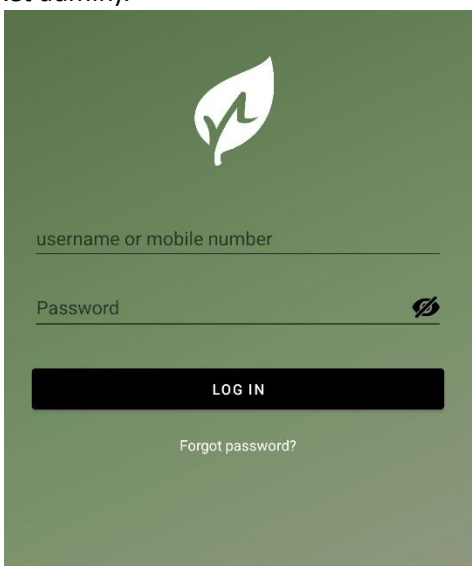
3. After confirming your location, you will be taken directly to the **Main Dashboard** to view available products and prices.

B. Seller Flow (I'M A SELLER)

If you select **I'M A SELLER**, the app requires verification of your seller account before granting access to seller tools:

1. Tap the **I'M A SELLER** (Teal) button.

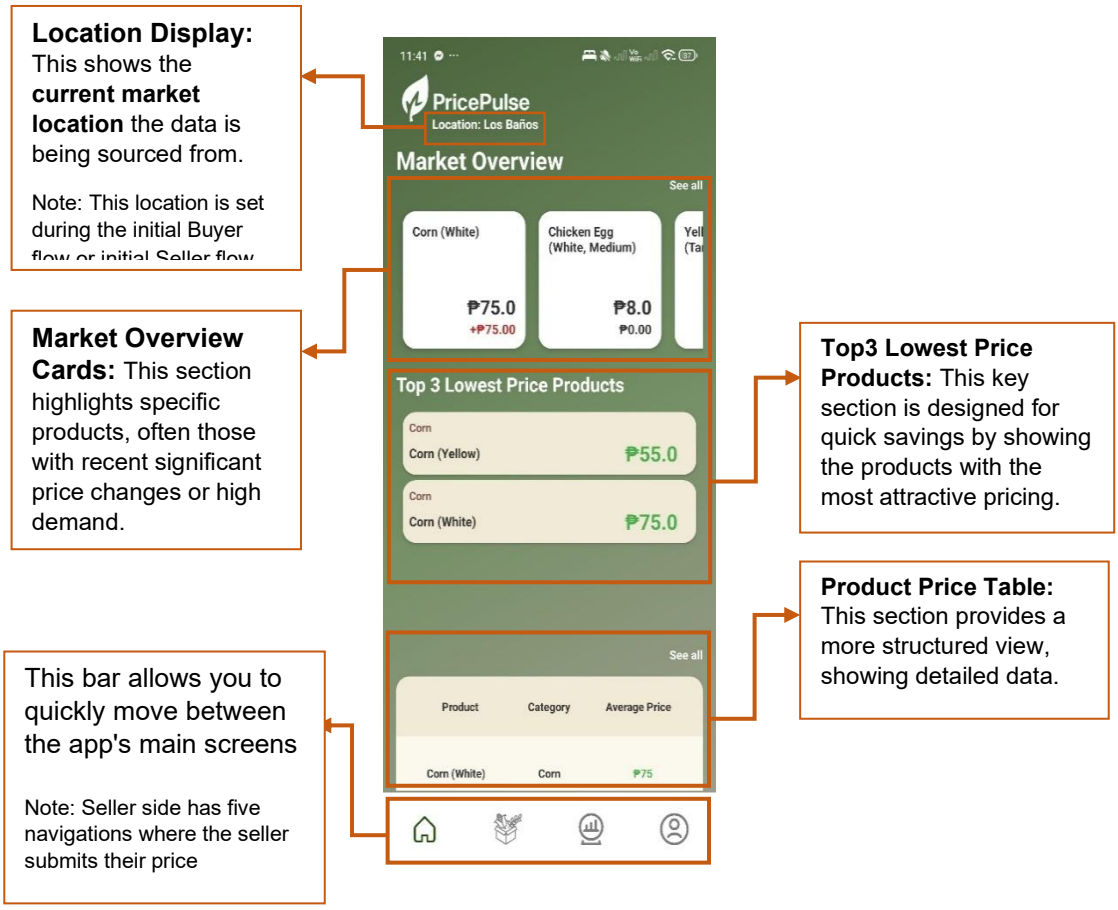
2. You will be directed to the **Login** screen where you must enter your credentials (Provided by the market admin).



3. Upon successful login, you will be taken directly to the **Main Dashboard** to view available products and prices.

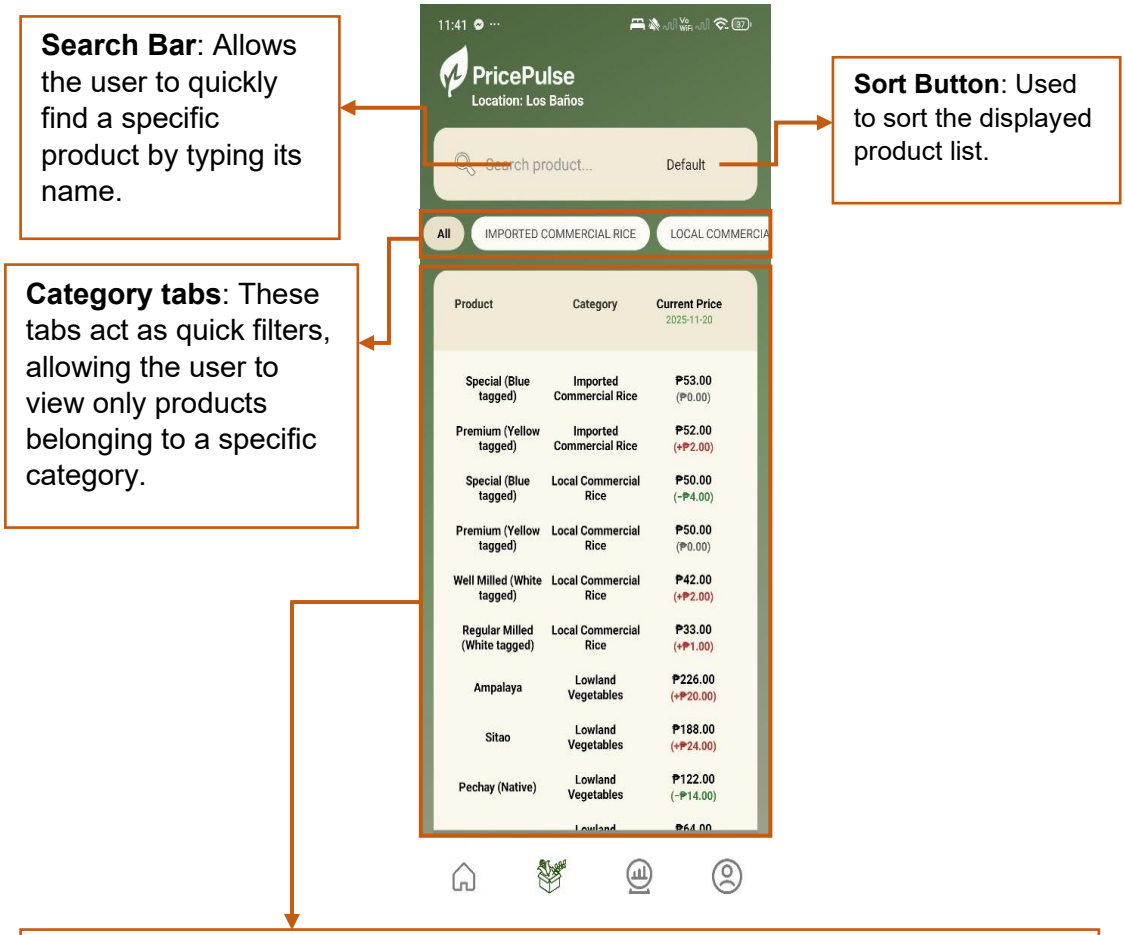
Main Dashboard Overview

The Dashboard provides a comprehensive **Market Overview** after you have selected your location or successfully logged in.



Commodities: Detailed Pricing and Hisorical Data

This screen provides a comprehensive table of all tracked products, their categories, and current prices in the market.



Search Bar: Allows the user to quickly find a specific product by typing its name.

Sort Button: Used to sort the displayed product list.

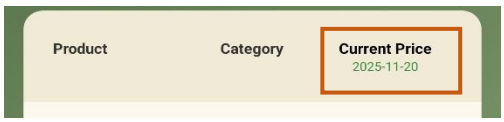
Category tabs: These tabs act as quick filters, allowing the user to view only products belonging to a specific category.

Price Table: The core element of the screen, displaying the following data columns:

- **Product**: The specific name and tag of the commodity (e.g., Special (Blue tagged)).
- **Category**: The broad grouping the product belongs to (e.g., Imported Commercial Rice).
- **Current Price**: The price recorded for the date listed. This also shows the

To view previous pricing:

1. Locate the date displayed in the Current Price column header (e.g., 2025-11-20).



2. Tap on the date within the column header.



3. A calendar or date selector should appear, allowing you to choose a previous date.

4. Once a new date is selected, the prices in the Current Price column will automatically update to reflect the market prices recorded on that historical date.

Forecast Price: Price Predictions

This screen provides predicted prices for various commodities based on data modeling and historical trends. This helps users anticipate future pricing for budgeting and procurement decisions.



Search Bar: Allows the user to quickly find a specific product by typing its name.

Category: These tabs act as quick filters, allowing the user to view only products belonging to a specific

Date Display (e.g., 2025-11-23): This shows the specific future date for which the prices in the table are currently being

Forecast Price Table: The central table providing the predictions:

- **Product**: The specific name and type of the commodity (e.g., Corn (White)).
- **Category**: The broad grouping the product belongs to (e.g., Corn).
- **Forecast Price**: The predicted price for the item on the selected

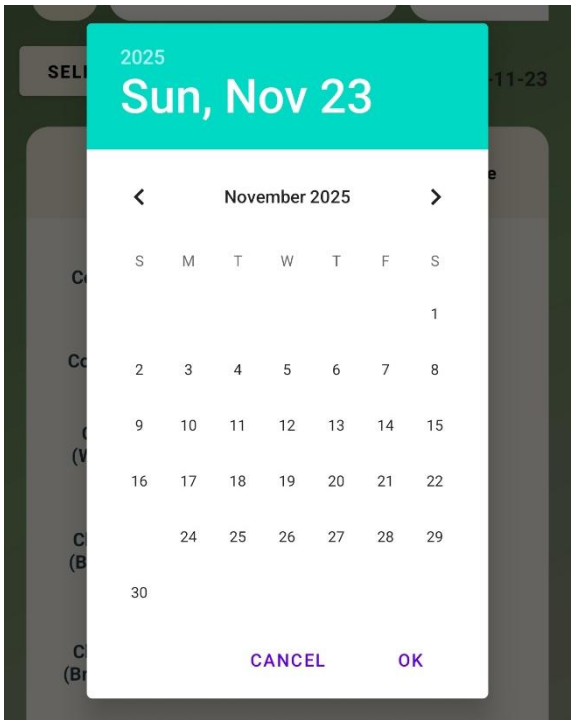
To view the predicted prices for a date other than the one currently displayed:

1. Locate the SELECT DATE button (next to the date display).



2. Tap the SELECT DATE button.

3. A calendar or date selector will appear, allowing you to choose a future

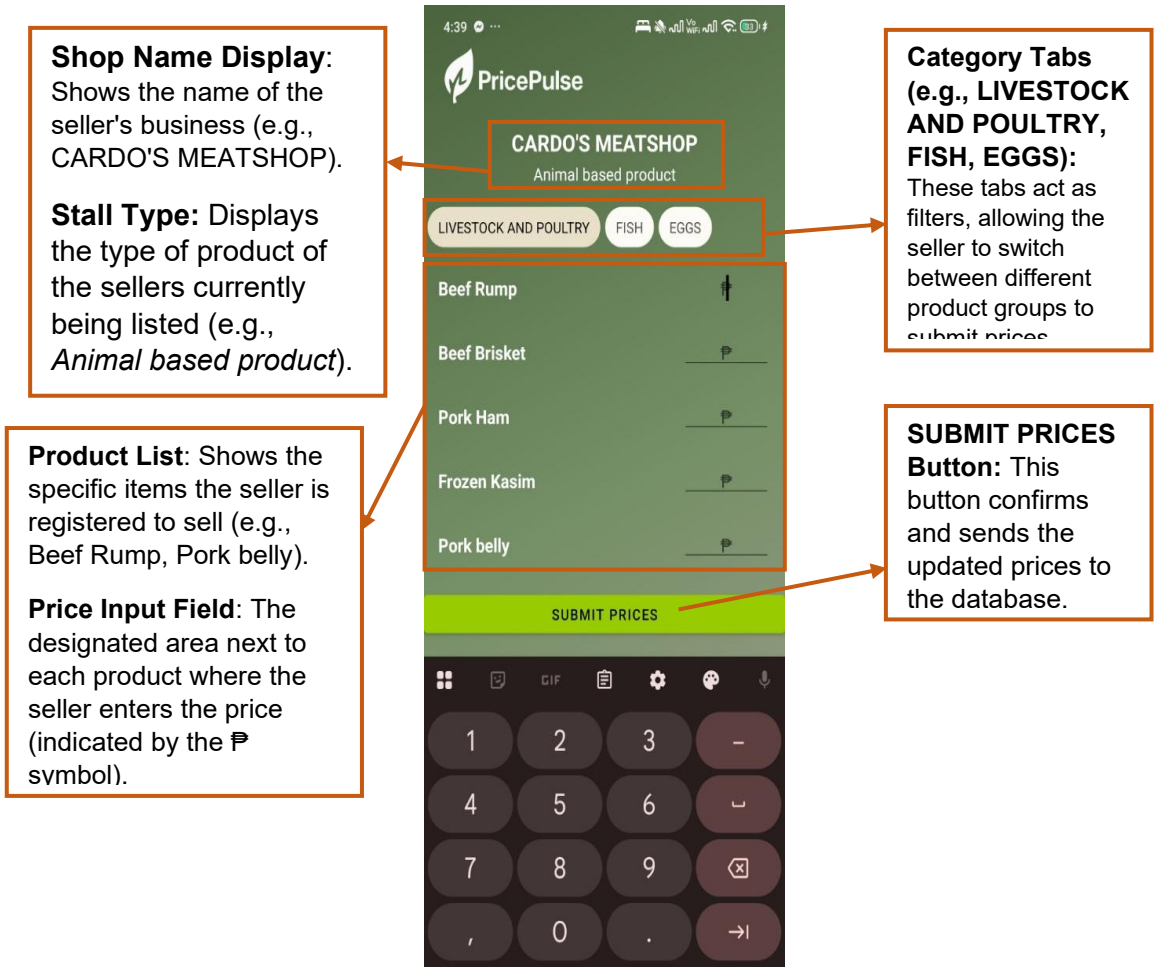


date.

4. Once a new future date is selected, the Forecast Price column will automatically update to show the predicted prices for that specific date.

Price Submission Screen (For Sellers Only)

This screen allows registered Sellers to enter and update the current selling price for each product they offer in the market.



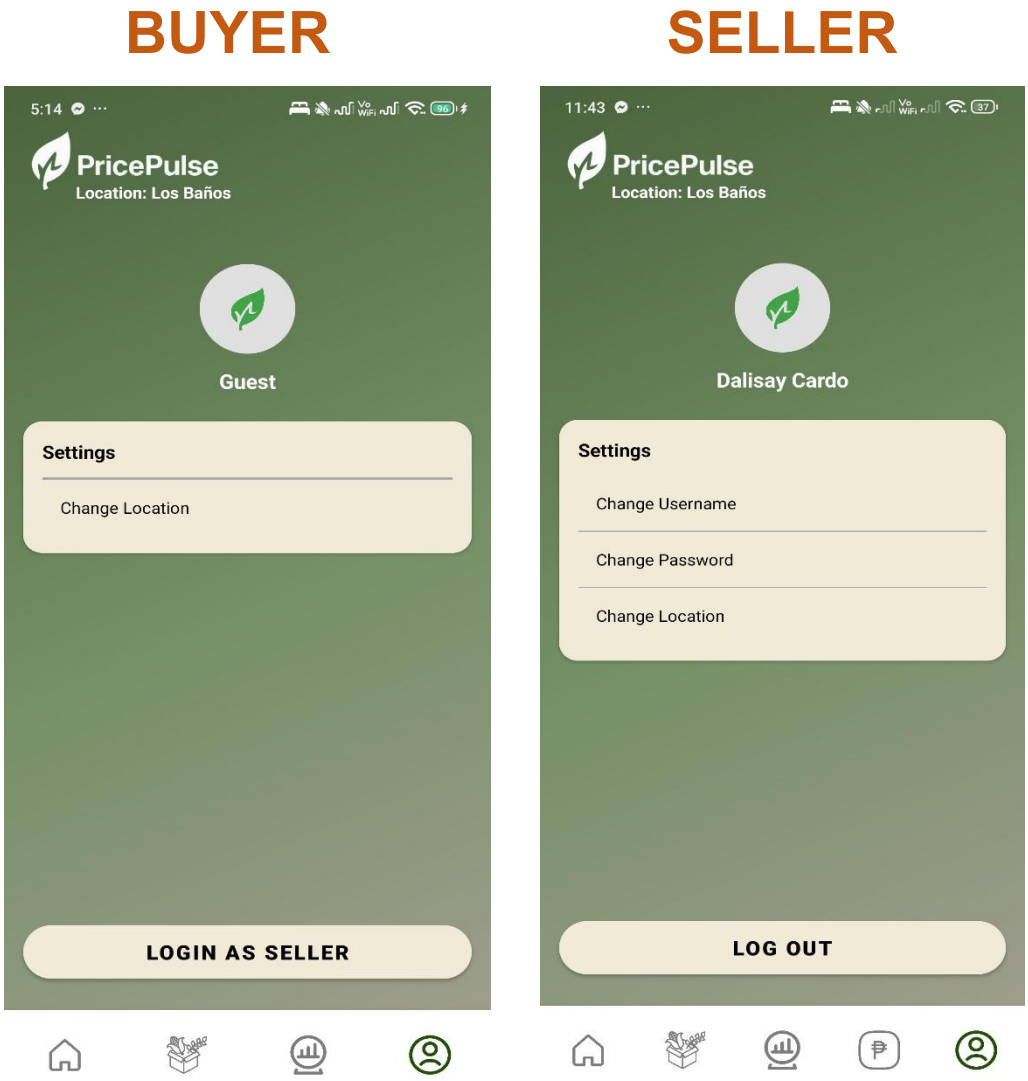
Submitting Prices (How It Works)

To successfully submit or update your product prices:

1. **Select Category:** Tap the relevant **Category Tab** (e.g., **FISH**) to display the list of products you wish to price.
2. **Enter Price:** Tap on the input field (marked with ₱) next to the specific product you wish to price. This action will open the on-screen numeric keyboard.
3. **Input Value:** Use the keyboard to enter the price you are currently selling the product for.
4. **Repeat:** Repeat steps 2 and 3 for all products you want to update within that category.
5. **Submit Prices:** Once you have finished entering your prices, tap the **SUBMIT PRICES** button located directly above the keyboard to finalize and send your prices to the market.

Profile Screen: Account and Settings

This screen, accessed via the person icon in the navigation bar, allows users to manage their account information, settings, and location. The options displayed vary based on whether the user is logged in (Seller) or is a



Guest/Buyer.

A. Logged-in User (Seller)

If you are logged in, you have full control over your account settings:

1. To change your displayed market data location: Tap Change Location to select a different area for viewing prices.
2. To update your login details: Tap Change Username or Change Password.
3. To exit your account: Tap the LOG OUT button.

B. Guest/Buyer User

If you are using the app as a Guest or Buyer, your primary options are limited:

1. To change your displayed market data location: Tap Change Location to select a different area for viewing prices.

2. To switch to a Seller role: Tap LOGIN AS SELLER to navigate to the login screen.

Appendix G

Publishable Paper (IEEE Format)

REAL-TIME PRICE MONITORING AND MARKET PREDITION FOR AGRICULTURAL COMMODITIES IN DISTRICT 2: A MULTI-PLATFORM DATA ANALYTICS APPLICATION

John Vincent V. Sorino
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jilianmarsierra@gmail.com

Abstract - This study addresses the persistent issue of price volatility and limited real-time access to agricultural commodity prices in District 2 of Laguna. To mitigate inflation-driven challenges and unfair market pricing, the project developed a multi-platform application—Price Pulse—that enables real-time monitoring and prediction of agricultural and poultry product prices. Using Agile-Scrum methodology, the system integrates an Android mobile app for consumers and sellers, and a web-based platform for administrators who validate submitted prices and compute market averages. Machine learning algorithms were applied to analyze historical data and generate short-term price predictions. System performance was evaluated using ISO software quality standards, covering functional suitability, usability, efficiency, and security. Results indicated "High" to "Very High" ratings across all criteria, demonstrating the system's reliability and accuracy. The findings suggest that Price Pulse can support fair trade, enhance market transparency, and strengthen local economic resilience.

Keywords: *Real-Time Price Monitoring, Agricultural Commodities, Market Prediction, Multi-Platform Application, Machine Learning.*

I. INTRODUCTION

Project Context

The Philippines has been experiencing persistent inflation, with soaring prices of essential goods, particularly food and fuel, which exacerbate the economic challenges faced by households. Key agricultural and poultry commodities, such as rice, vegetables, onions, tomatoes, and poultry, have been significantly affected, impacting both consumers and local farmers. The inflation rate often surpasses wage growth, compelling families to reduce spending, which has downstream effects on local markets and supply chains.

In Laguna, a province characterized by a mix of urbanization and agriculture, local markets experience fluctuating prices that directly influence both producers and consumers. Agricultural livelihoods are especially vulnerable to environmental calamities, such as typhoons and droughts, as well as inadequate storage and distribution systems, leading to seasonal price spikes and market instability. Informal trade practices and inconsistent information flow further contribute to unfair pricing, widening economic inequalities in agricultural communities.

Traditional price monitoring in the Philippines relies on periodic surveys conducted by agencies such as the Philippine Statistics Authority (PSA) and the Department of Agriculture (DA). However, these reports are often delayed and may not reflect real-time conditions. While time series analysis methods such as ARIMA have been utilized in research for price prediction, they have yet to be widely adopted by local government units or small-scale farming operations.

These challenges highlight the need for a localized, real-time price monitoring system that integrates market data, enhances decision-making, and provides predictive analytics. This study proposes the development of a multi-platform, machine learning-powered application for real-time price monitoring and market prediction in Laguna's Second District, which includes Cabuyao City and the municipalities of Bay and Los Baños. The proposed system bridge information gaps, reduce price volatility, support fair trade practices, and promote a more resilient local agricultural economy.

Project Objectives

The primary objective of this project is to develop a multi-platform application that provides real-time monitoring and market price prediction for agricultural and poultry products in Laguna's Second District using data analytics. The specific objectives are as follows:

1. Application Development: Develop a web-based and mobile-based application that delivers real-time prices of agricultural and poultry products to end-users, including farmers, vendors, and consumers.

2. Data Analytics Integration: Implement analytics features that visualize price trends, predict market fluctuations, and provide actionable insights to improve decision-making in agricultural trade.
3. System Evaluation: Assess the performance of the developed system in terms of accuracy, usability, and reliability through user testing, feedback collection, and comparison with existing price monitoring methods, following ISO standards.

II. RELATED LITERATURE

A. Analysis and Impact of Price Volatility

Price volatility in agricultural commodities is influenced by factors such as seasonal changes, supply-demand imbalances, and external shocks. Piotr et al. [1] analyzed data from the Ministry of Agriculture and Rural Development in Poland and found that global crises contributed to significant price fluctuations, which negatively affected both consumers and producers. Kemény et al. [2] similarly emphasized that price volatility can lead to welfare losses and challenges in production planning.

Various models have been applied to analyze price volatility. Khabibah et al. [3] employed ARCH/GARCH models to study big cayenne prices in Malang District, Indonesia, finding low volatility at both producer and consumer levels. Calvo [4] demonstrated that sudden stops increase relative price volatility, using ARCH techniques to analyze the wholesale-to-consumer price ratio. Supriya and Mamilla [5] highlighted the importance of incorporating seasonality into price predictions using GARCH models, improving the accuracy of forecasting market movements.

B. Application of Mobile and ICT Solutions in Agricultural Markets

Mobile applications and ICT tools have increasingly been used to enhance market transparency and empower smallholder farmers. Kamal and Bablu [6] discussed how mobile platforms provide real-time access to market information, weather updates, and best farming practices, while facilitating direct connections between farmers and buyers to reduce intermediaries and improve price transparency.

Rajon et al. [7] presented an SMS-based ICT application for disseminating commodity price information, enabling consumers and sellers to transact within government-defined fair price margins. Charleboi et al. [8] noted that retail price volatility directly influences consumer purchasing behavior, emphasizing the need for reliable price monitoring. Li et al. [9] applied a hybrid Particle Swarm Optimization-Genetic Algorithm (PSO-GA) model to forecast scallion prices in Shandong Province, demonstrating improved predictive accuracy.

Several Philippine-based initiatives also highlight ICT applications for market monitoring. Cayabyab et al. [10] developed *Price Watch*, a mobile app providing real-time price updates of agricultural products in public markets using Exponential Smoothing forecasting. Fafchamps and Aker [11] demonstrated that mobile-based price reporting improves transparency and allows farmers to bypass intermediaries. Bista et al. [12] introduced *Crop Mandi*, an app leveraging data analytics to forecast crop demand and prices, helping farmers plan production to maximize profitability.

C. Producer Price Index and Agricultural Market Trends

Monitoring the Producer Price Index (PPI) is critical for understanding market trends. Del Prado [13] reported fluctuations in the Philippine PPI for crops, livestock, and poultry between 2023–2024, reflecting seasonal, supply, and demand dynamics. Studies such as Pang et al. [14] and Gomez [15] emphasized that factors like epidemic outbreaks, input costs, and government policies significantly influence pork and livestock prices. Price monitoring helps stakeholders mitigate losses and make informed production and trading decisions.

Domestic studies highlight that efficient animal production systems, supported by ICT and predictive analytics, can improve household income, food security, and market resilience. Domingo et al. [16] and Briones and Espineli [17] underscore the role of industry benchmarking and economies of scale in livestock, poultry, and dairy production. Devendra and Chantalakhana [18] also stress the importance of natural resource management and technology adoption in reducing rural poverty.

III. METHODOLOGY

Project Design

Figure 1 illustrates the workflow of a Price Monitoring System. A Market Officer/Admin in specific locations (Cabuyao, Bay, Los Baños) initiates the process by collecting prices from a Seller. The Seller then submits these collected prices to the Price Monitoring System and also posts prices/approves data back to the Market Officer. The Price Monitoring System serves as the central hub: it sends the price data to a separate Analytics and Prediction module.

This module processes the data and returns the resulting analytics and predictions back to the system.
Finally, the Price Monitoring System displays the prices along with the derived analytics and predictions to the Public Users.

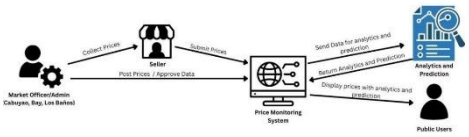


Figure 1. System Architecture

Project Development

Figure 2 illustrates the Scrum Framework, a popular methodology within Agile project management. The process begins with the Product Owner maintaining the Product Backlog, which is a prioritized list of features and requirements. The Team then collaborates in the Sprint Planning Meeting to select items from the Product Backlog, creating the Sprint Backlog. This Sprint Backlog is the work to be completed during a Sprint, which typically lasts for one to three weeks. During the Sprint, the Scrum Master facilitates Daily Scrums (also known as Daily Stand-ups) to synchronize the team's activities and remove impediments. This iterative cycle of Sprints moves the project forward, ultimately resulting in Finished Work (a potentially shippable increment of the product) at the end of each Sprint.

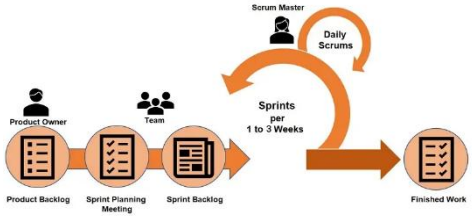


Figure 2. Agile-Scrum Methodology

Evaluation Procedure

The evaluation procedure for the system comprehensively assess its effectiveness across four key areas, utilizing real users from the Los Baños Public Market. Functional Testing verify that all components, the mobile application, admin panel, and prediction model, operate according to their intended design, using both manual and automated tests for critical features like price submission and user validation. Usability Testing measure the system's ease of use for farmers, vendors, and administrators through structured interviews, feedback sessions, and the quantifiable measure of user satisfaction via a System Usability Scale (SUS) survey. The Evaluation of Prediction Accuracy gauge the reliability of the forecasting algorithm by comparing its predicted values against actual historical data using key metrics such as Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE). Finally, Performance Testing assess the system's speed and reliability by simulating concurrent user activity to measure load times, latencies, and the synchronization efficiency of real-time data updates.

Evaluation Procedure

The study's population consists of key stakeholders involved in agricultural and poultry trading within the public markets of Cabuyao City, Bay, and Los Baños, all located in District 2 of Laguna. The primary participant groups are Vendors/Sellers who set prices, Consumers who are regular market buyers, and Market Administrators who manage market operations and pricing data. A purposive sampling technique employed to select an approximate total of 15 to 60 respondents. This targeted selection ensures a balanced representation from each stakeholder group to facilitate comprehensive data collection and gather essential user feedback for the system's development and subsequent evaluation.

Statistical Treatment of Data

The statistical treatment involves analyzing data collected from the target respondent pool using a combination of population and purposive sampling techniques to comprehensively evaluate the system's functionality and performance. This dual sampling approach ensures a representative sample and facilitates a thorough examination of diverse user perspectives, which in turn informs a tailored, user-centric development process. To gauge user satisfaction with the application's features, developers utilized a **Likert Scale** method, where respondents provided one of five impressions, ranging from five (Very High) to one (Very Low). The collected survey feedback were then be processed using a percentage-based evaluation method for detailed analysis. The Likert Scale defines the scoring: a Mean Range of $4.01 – 5.00$ is interpreted as Very High, $3.25 - 4.0$ as High, $2.50 - 3.24$ as Moderate, $1.75 - 2.49$ as Low, and $1.0 - 1.74$ as Very Low.

IV. RESULT AND DISCUSSION

Evaluate the system's performance in terms of accuracy, usability, and reliability through user testing, feedback collection, and comparative analysis with existing price monitoring methods using ISO testing.

The system was rigorously evaluated using the ISO 25010 software quality framework, and results showed consistently "Very High" ratings across all measured quality attributes. Functional Sustainability recorded the highest score at 4.88 ±0.33, reflecting the system's strong ability to perform intended functions accurately and consistently. High ratings for Security and Portability (both 4.80 ±0.41) demonstrate the system's capability to protect user data and operate smoothly across different platforms. In addition, the system exhibited strong Usability (4.70 ±0.46) and Compatibility (4.66 ±0.51), indicating that users found the interface intuitive and the system well-suited for integration with various devices and environments.

Items	Mean	SD	Interpretation
Functional Sustainability	4.88	±0.33	Very High
Performance Efficiency	4.33	±0.50	Very High
Compatibility	4.66	±0.51	Very High
Usability	4.70	±0.46	Very High
Readability	4.50	±0.51	Very High
Security	4.80	±0.41	Very High
Maintainability	4.41	±0.51	Very High
Portability	4.80	±0.41	Very High
Composite Mean	**4.63**	±0.45	**Very High**

n=3, Legend: highest to lowest as follows: five (5) – Very High (4.21 – 5.00), four (4) – High (3.41 – 4.20), three (3) – Moderate (2.61 – 3.40), two (2) – Low (1.81 – 2.60), one (1) – Very Low (1.00 – 1.80).

Table 8. ISO 25010 Testing

Other evaluated aspects—including Performance Efficiency (4.33 ±0.50), Readability (4.50 ±0.51), and Maintainability (4.41 ±0.51)—also received "Very High" interpretations, affirming that the system is fast, easy to understand, and simple to update or enhance. With an overall composite mean of 4.63 ±0.45, the findings confirm that the system meets high-quality software standards and performs reliably, making it well-suited for real-time agricultural and poultry price monitoring and long-term operational use.

Table 8 reveals a consistently positive user perceptions of the price pulse app with an overall composite mean of 4.42, indicating strong agreement. The result shows that the app helps users perform market price tracking efficiently and accurately, as evidenced by a mean rating of 4.40. The respondents also perceived the app as valuable in providing useful data insights for market-related decision-making, which received one of the highest ratings at 4.53. Moreover, the app was viewed as contributing to improve work productivity, with an average of 4.35. the accuracy and reliability of the information generated by the system were also rated very high in two separate terms items, with an average score of 4.38 and 4.55, that indicates a strong confidence in the system's data quality. Overall, the composite mean of 4.42 with a standard deviation of ±0.63 demonstrates that respondents consistently perceived the app as highly useful, dependable, and beneficial for market monitoring and decision-making tasks.

V. CONCLUSION AND RECOMMENDATION

The development of Price Pulse, a multi-platform web and mobile system, represents a significant technological improvement for the Department of Agriculture, particularly in monitoring agricultural and poultry product prices within Laguna's District 2. The system successfully achieved its primary objective by providing real-time price tracking through a multi-platform application that incorporates data entry, administrative verification, and data visualization. This functionality ensures the availability of precise and current pricing information for vendors and consumers, thereby promoting fair trading and informed decision-making. Price Pulse effectively integrates data analytics features by utilizing charts, graphs, and dashboards to visualize price trends, track seasonal variations, and display historical and present prices. Crucially, the system incorporates predictive models based on data analysis to anticipate market fluctuations, enabling stakeholders to better plan production, sales, and inventory, thus optimizing earnings and increasing market competitiveness.

Finally, a comprehensive evaluation confirmed the system's efficacy: user testing demonstrated a high degree of accuracy in price predictions with only minor deviations, user feedback rated the design as easy to use and straightforward, and the system proved to be more reliable and effective at updating and monitoring data compared to existing methods. In conclusion, the Price Pulse system successfully meets its objectives, providing a valuable, technologically advanced tool for real-time price monitoring and assessment.

Recommendations

The recommendations for the future development of the Price Pulse system focus on enhancing its prediction capabilities, administrative functionality, data integrity, and compliance.

1. Prediction Enhancement: Future research should integrate more advanced and complex predictive models, specifically adopting Artificial Intelligence (AI) techniques like machine learning or deep learning to analyze larger datasets, uncover hidden patterns, and generate significantly more precise price predictions.

2. External Factors: The forecasting framework should be made more adaptive and comprehensive by incorporating AI-driven measures that account for external environmental and biological variables, such as natural disasters, animal diseases, and crop destruction from pest infestations, to better reflect real-world agricultural conditions.

3. Administrative Mobility: It is recommended to create a dedicated mobile application for the system administrators (admin). This innovation allows municipal administrators to monitor and operate the system remotely, even when away from the office or at home.

4. Data Integrity and User Engagement: Feedback or reporting component should be added to enable users to report suspicious or incorrect price entries. This feature foster a cooperative environment, enhance data validation, integrity, and transparency while help to effectively detect mistakes or potential price manipulation.

5. Official Compliance: To ensure the system's long-term viability and potential for mass public deployment as an official government tool, it must be upgraded to comply with the Department of Agriculture (DA) standards. This includes aligning data structures with the DA-Bantay Presyo standard and ensuring standardized reporting formats and interoperability.

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APPENDIX H

Grammarly Result

REAL-TIME PRICE MONITORING AND MARKET PREDICTION FOR AGRICULTURAL COMMODITIES IN DISTRICT 2: A MULTI-PLATFORM DATA ANALYTICS APPLICATION

by Ronah joy Ramirez

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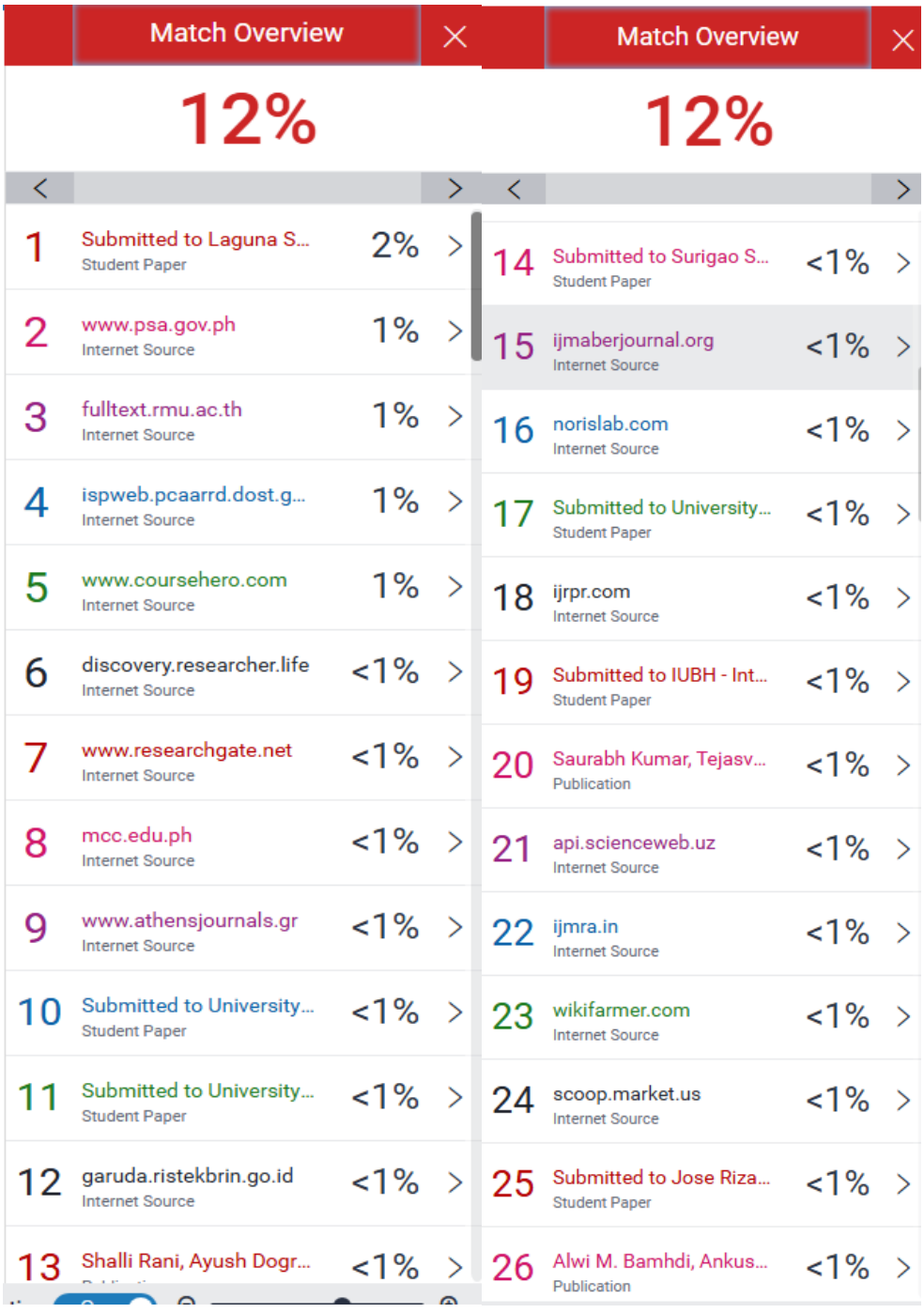
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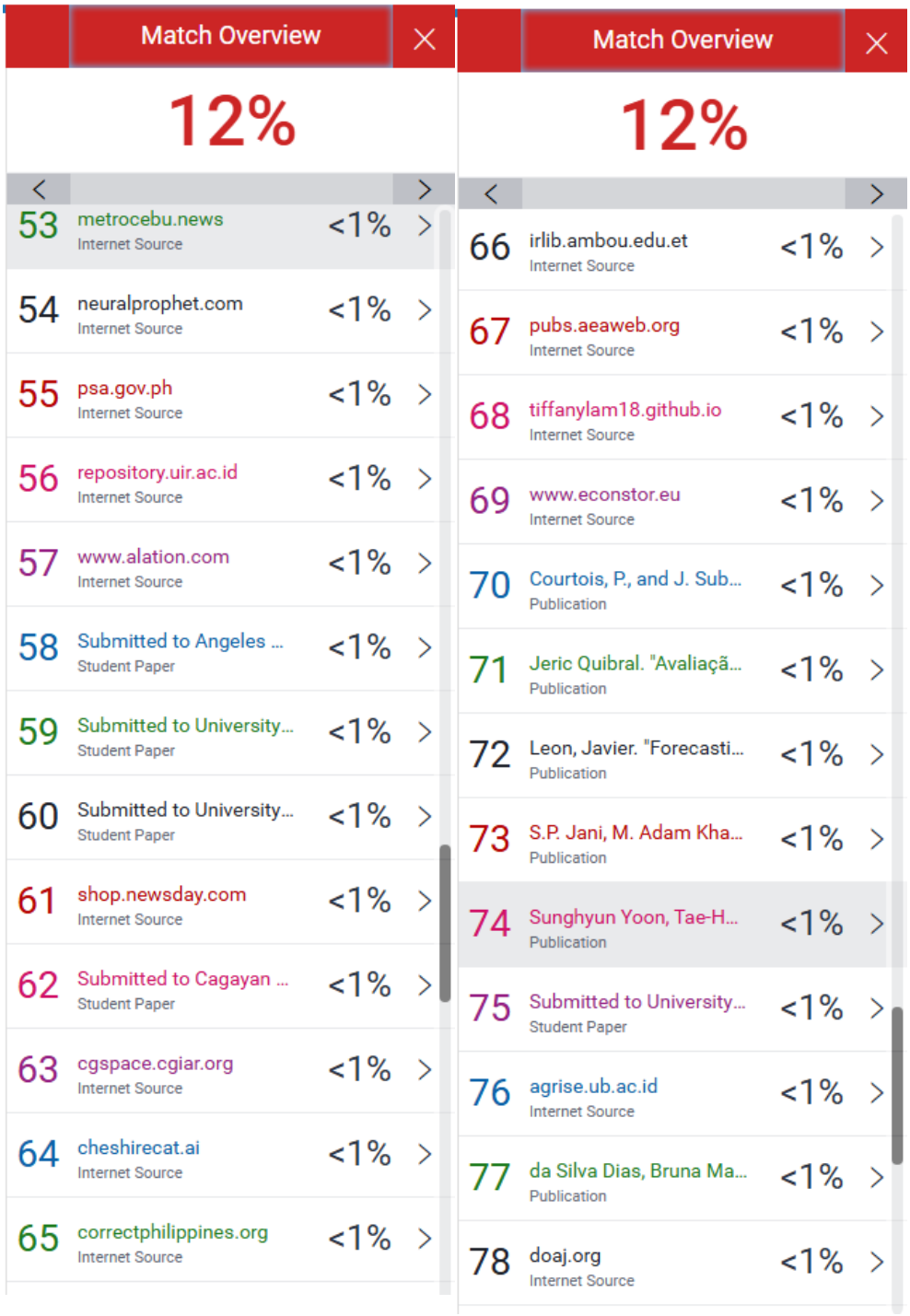
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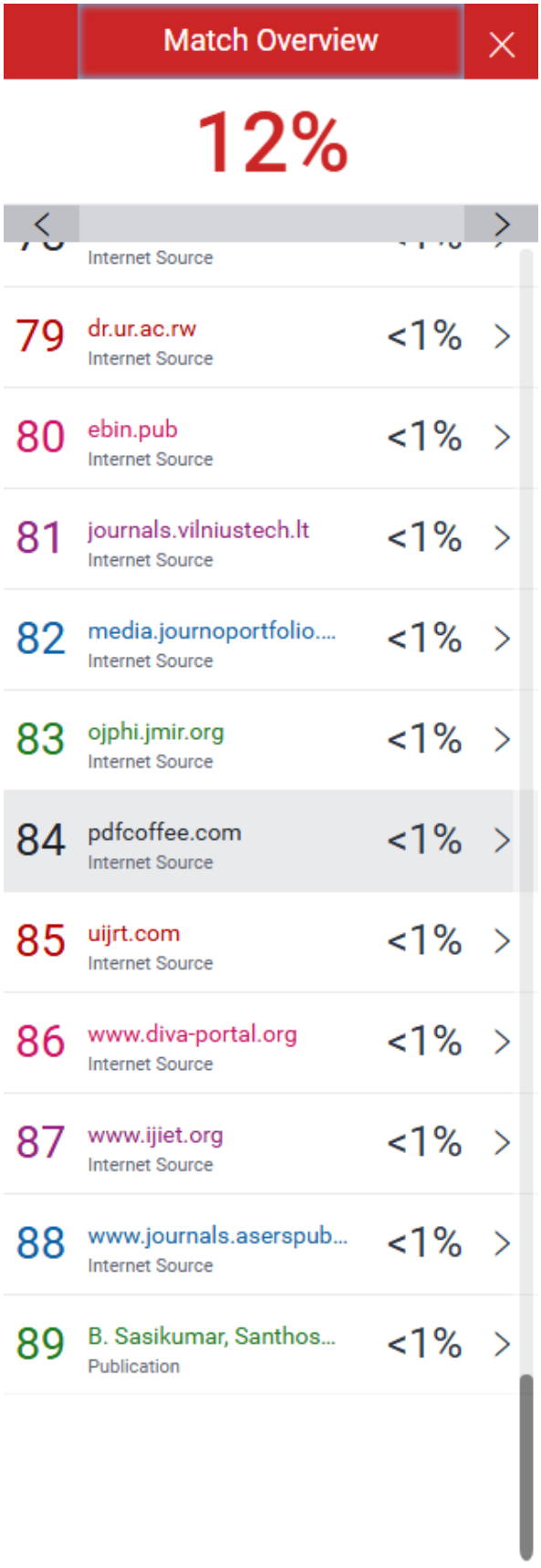
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